Integrating AI models into Rhino 8 STF Workshop:

Transforming Design Workflows with Computer vision Techniques

PRESENTERS |
IVANA PETRUSEVSKI - ESTHER RUBIO MADROÑAL - HESHAM SHAWQY - EVE TOBEY - CALLUM SYKES

GRIMSHAW
DESIGN
TECHNOLOGY

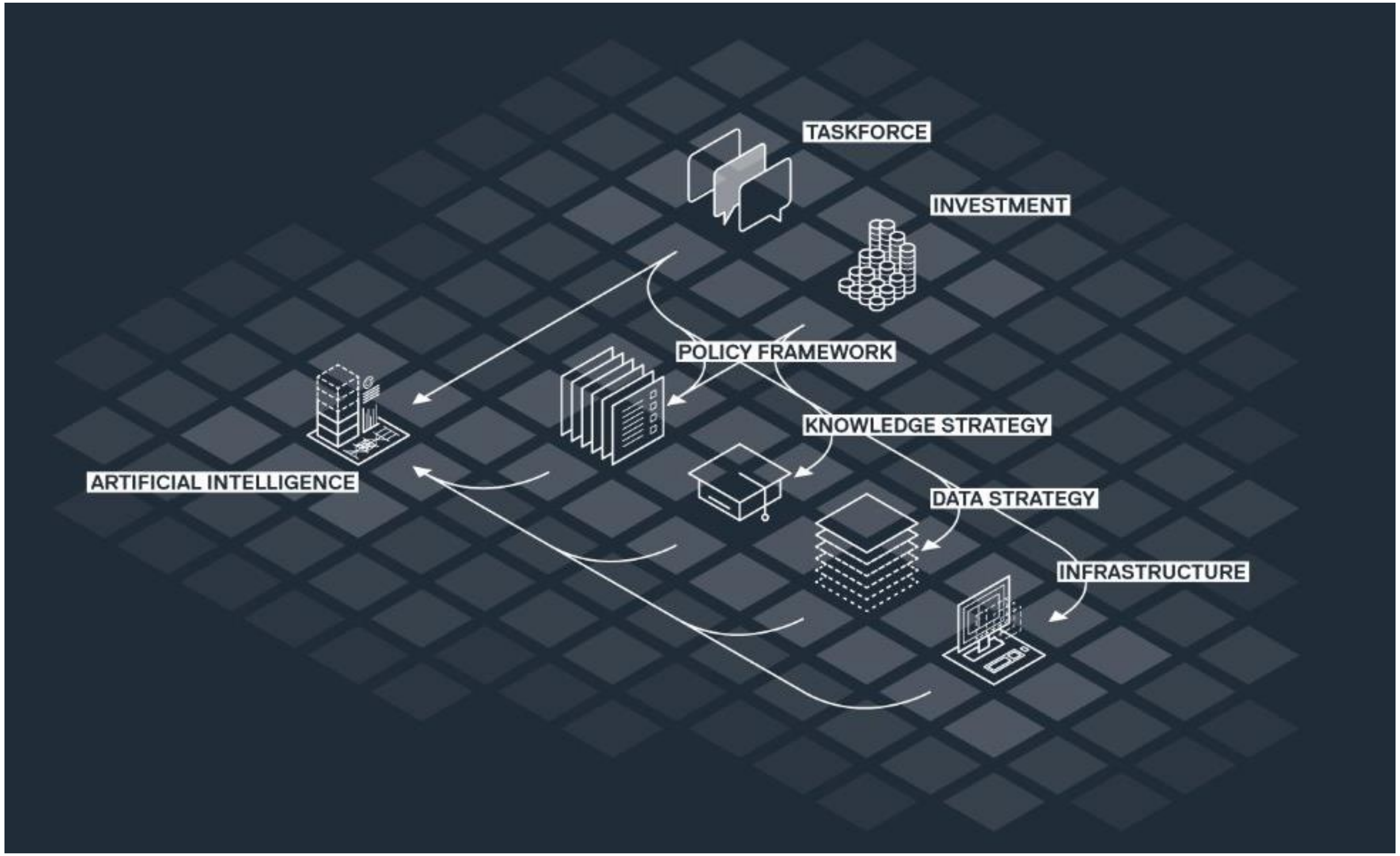
INTRODUCTION

GRIMSHAW
DESIGN
TECHNOLOGY

AI in Grimshaw
Design Technology

At GRIMSHAW, the AI working group focussed on AI through 6 different lenses:

- AI for design
- AI for operations
- Value of AI
- AI governance
- AI infrastructure
- Engagement



Introduction
Grimshaw Design Technology & McNeel

IVANA PETRUSEVSKI
Computational Design Lead, EMEA
Grimshaw Architects
(LONDON)

HESHAM SHAWQY
Computational Design Specialist
Grimshaw Architects (LONDON)

ESTHER RUBIO MADROÑAL
Senior Computational Design Specialist
Grimshaw Architects
(LONDON)

EVE TOBEY
Computational Design Specialist
Grimshaw Architects
(NEW YORK)

CALLUM SYKES
Software Developer
McNeel

Workshop Objectives:

- Introduce participants to computer vision and its applications in architectural design.
- Explore the integration of open-source AI models with Rhino 8.
- Demonstrate AI-driven techniques for design automation and visual analysis.
- Empower participants to leverage AI to enhance computational design workflows.

Expected Workshop Outcomes:

- Build AI-powered workflows using Python within Rhino8’s new Script Editor.
- Utilize a library of computer vision components or commands for tasks such as image segmentation, object recognition, and visual pattern analysis.

09:30 — 30 mins
1-Introduction | 4

- Round of introductions — Presentation — Hesham Shawqy
- Computer Vision overview — Presentation — Hesham Shawqy
- Workshop agenda — Presentation
- Workshop files — Presentation

10:00 — 30 mins
2-Setup | 3

- Rhino 8-GH-Visual Studio-Miniconda — Visual Studio
- Cuda - Virtual Environment — Visual Studio
- AI models checkpoints — Visual Studio

10:30 — 60 mins
Basic workflows | 4

- Text to image — Visual Studio — Hesham Shawqy
- Image to image — Visual Studio — Hesham Shawqy
- Controlnet to image — Visual Studio — Hesham Shawqy
- Scaling up the model — Visual Studio — Hesham Shawqy

11:30 — 30 mins
Short Break | 0

12:00 — 60 mins
Running in Rhino | 4

- Intro-Overview — Presentation — Callum Sykes
- viewcapture to simple processing — Rhino — Callum Sykes
- Viewcapture to AI image generation — Rhino — Callum Sykes
- Etoforms + AI image generation — Rhino — Callum Sykes

13:00 — 60 mins
Lunch Break | 0

14:00 — 60 mins
Running in GH | 3

- Viewcapture to AI image generation — Grasshopper — Esther Rubio Madronal
- Bulk AI image rendering — Grasshopper — Esther Rubio Madronal
- Form Finding + AI image generation — Grasshopper, Rhino — Esther Rubio Madronal

15:00 — 30 mins
Short Break | 0

15:30 — 60 mins
Extended Workflows | 5

- Intro-Overview — Presentation — Eve Tobey
- Image segmentation — Visual Studio — Eve Tobey
- Depth estimation — Visual Studio — Eve Tobey
- Object detection — Visual Studio — Eve Tobey
- AI-Streetviews — Rhino, Visual Studio — Hesham Shawqy

16:30 — 60 mins
Project | 2

- Project-Overview — Presentation — Ivana Petrusevski
- Working Session

17:30
Closing | 2

- Future Steps
- Thank you!

Workshop Schedule

1. Intro

// Presentation

- Introductions
- Computer Vision
- Workshop agenda

2. Setup

// Visual Studio

- Workshop files
- Rhino 8-GH-Visual
- Cuda - Virtual Environment
- AI models checkpoints

3. Basic Workflows

// Visual Studio

- Text to image
- Image to image
- Controlnet to image
- Scaling up the model

Break

4. Running in Rhino

// Rhino

- Intro-Overview
- Viewcapture to simple processing
- Viewcapture to AI image generation
- Etoforms + AI image generation

Lunch

5. Running in GH

// Rhino // Grasshopper

- Viewcapture to AI image generation
- Bulk AI image rendering
- Form Finding + AI image generation

Break

6. Extended Workflows

// Visual Studio // Rhino

- Intro-Overview
- Image segmentation
- Depth estimation
- Object detection
- AI-Streetviews

7. Project

// Hands on

- Plug and play

8. Closing

// Presentation

- Future Steps
- Thank you

```
S2F-AI-SCRIPTEDITOR
 ├── 1-Introduction
   └── intro.ppt
 ├── 2-Setup
   ├── ai_checkpoints.py
   └── virtual_environment.py
 ├── 3-Img_Img_workflows
   ├── 01-text_to_image.py
   ├── 02-image_to_image.py
   ├── 03-controlnet_to_image.py
   └── 04-model_scaleup.py
 ├── 4-Running_in_Rhino
   ├── 01-viewcapture_to_processing.py
   ├── 02-viewcapture_to_ai.py
   ├── 03-etoforms_to_ai.py
   └── 04-formfinding_to_ai.py
   └── intro.ppt
 ├── 5-Running_in_Grasshopper
   ├── 01-viewcapture_to_ai.gha
   └── 02-build_ai_rendering.gha
 ├── 6-Expanded_workflows
   ├── 01-Intro.ppt
   ├── 02-image_segmentation.py
   ├── 03-depth_estimation.py
   ├── 04-object_detection.py

└── README.md
```

GitHub

https://github.com/design-technology/S2F-AI-ScriptEditor/tree/Workshop

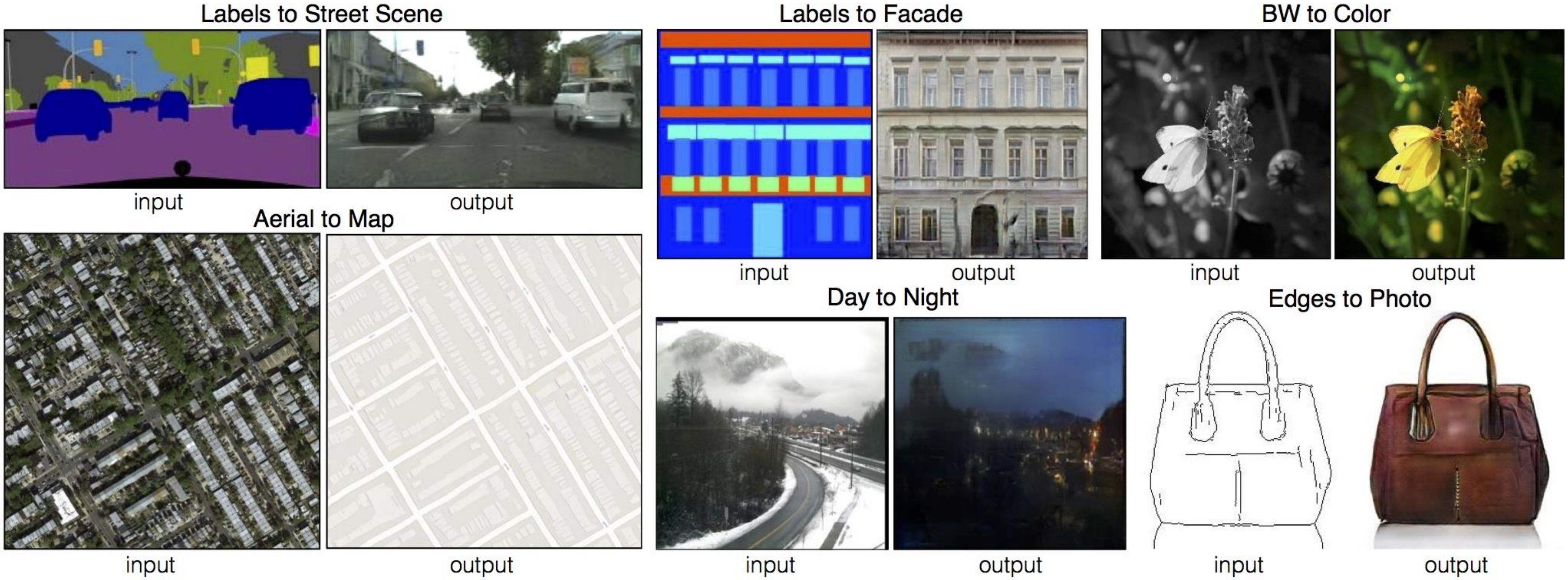
COMPUTER VISION
GRIMSHAW

Early Computer Vision Applications
GANs - 2020

https://github.com/junyanz/pytorch-CycleGAN-and-pix2pix

https://www.michaelhasey.com/gan-exterior

Early Computer Vision Applications
PIX2PIX - 2020



https://phillipi.github.io/pix2pix/

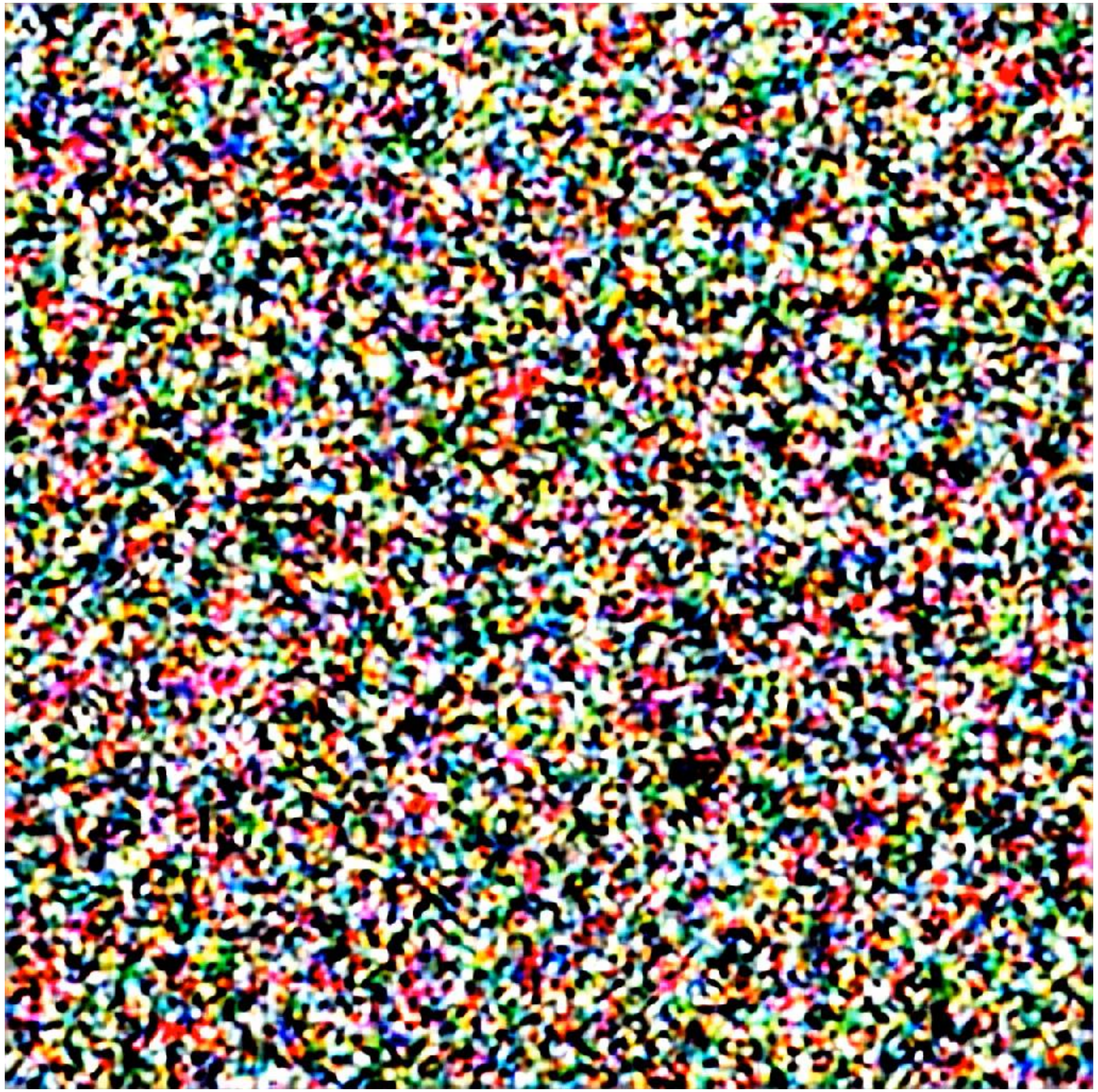
Diffusion Models
Scaling up, 2022

SCALING UP

Midjourney

OpenAI

stability.ai



Diffusion Models
Evolution - 2022:2023

a photograph of a residential multistory building with timber cladding and green landscaping, hyper realistic

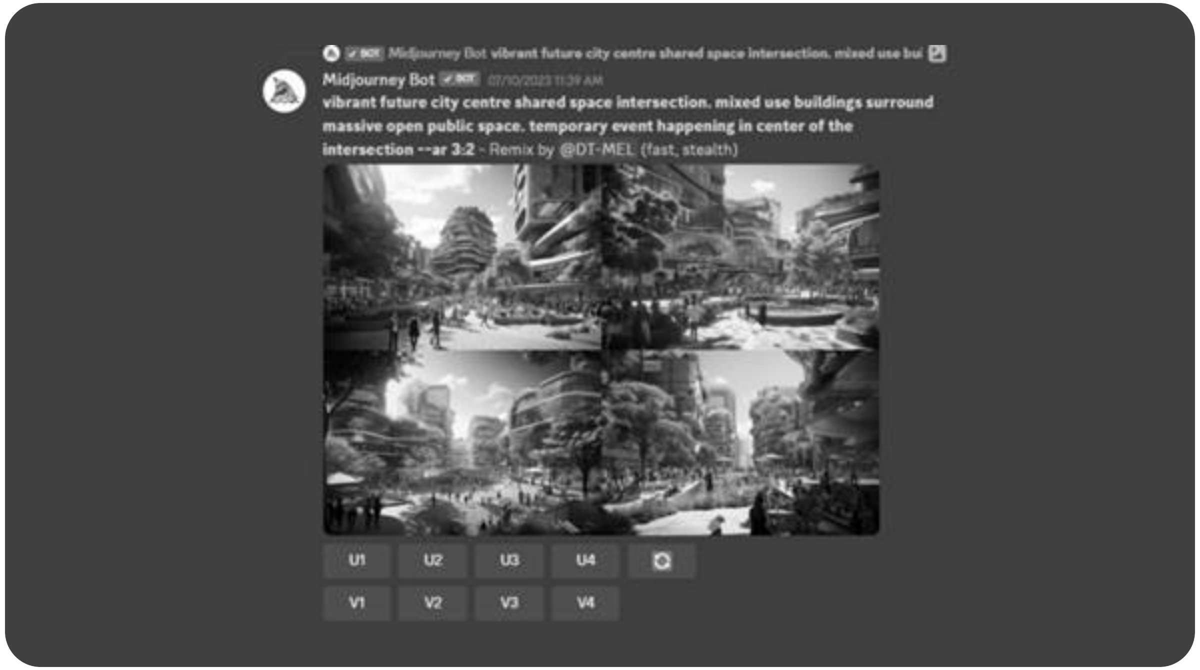
V1 – Feb 2022

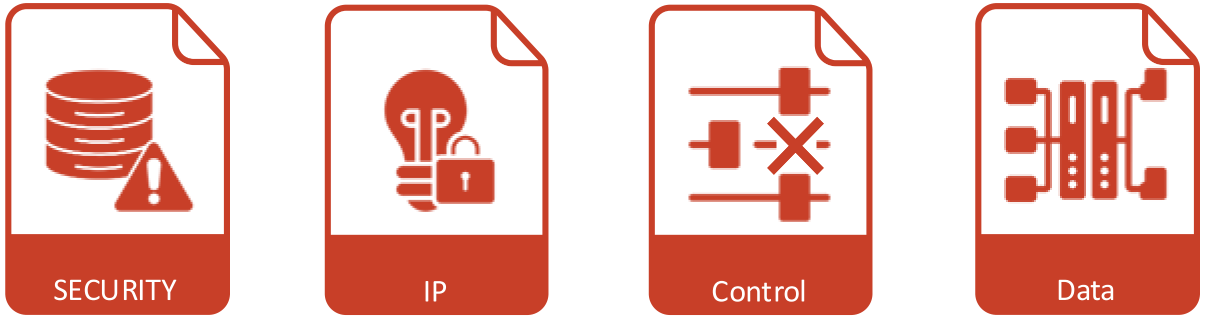
V6 – Dec 2023

Midjourney

AI Rendering Inhouse tool

Motive - Practice





Web based AI Rendering Inhouse tool
Framework

aws python™

Hugging Face

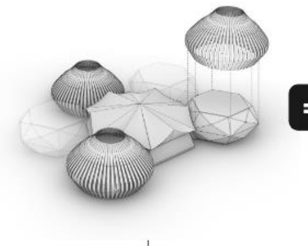
a stunning view of a **cluster of modular pavilions** nestled within the lush **Brazilian jungle**, the roof is built using woven bamboo elements, surrounded by majestic mountains rising in the background and **a serene river flowing in the foreground**, the **trees are way taller than the pavilions**, earthy tones that blend harmoniously with the **yellowish greens of the surrounding jungle**, **volumetric sunlight goes across the jungle**, creating fascinating **light rays**, 4k, high resolution, **realistic render**, architectural visualization

Long prompts - Prompt Weighting - Multi Generation - Image Preprocessing - Inpainting - Upscale - Sketch2image - Modelcapture2image - regions2image

+

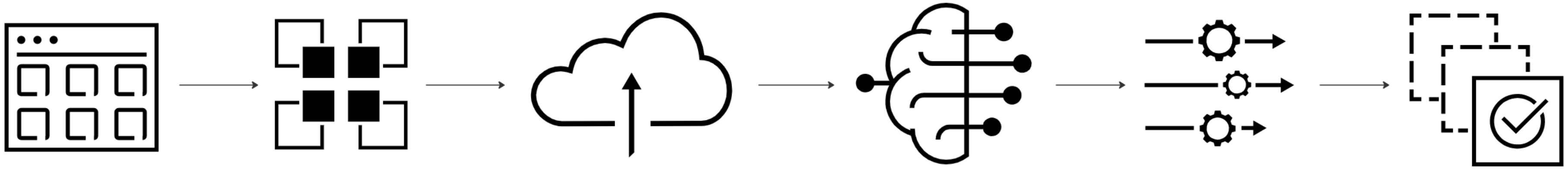
Geometry Weight
Prompt Weight
Number of Images
Iteration Seed

+

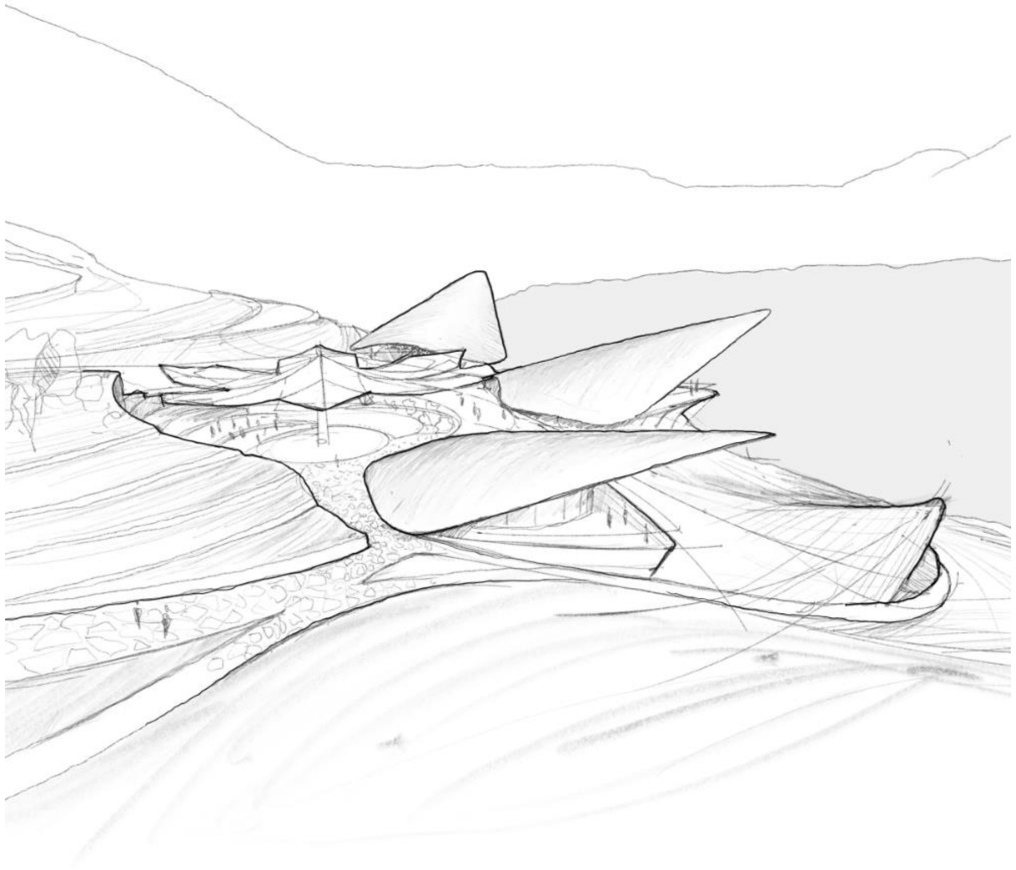


GEOMETRY LAYERS

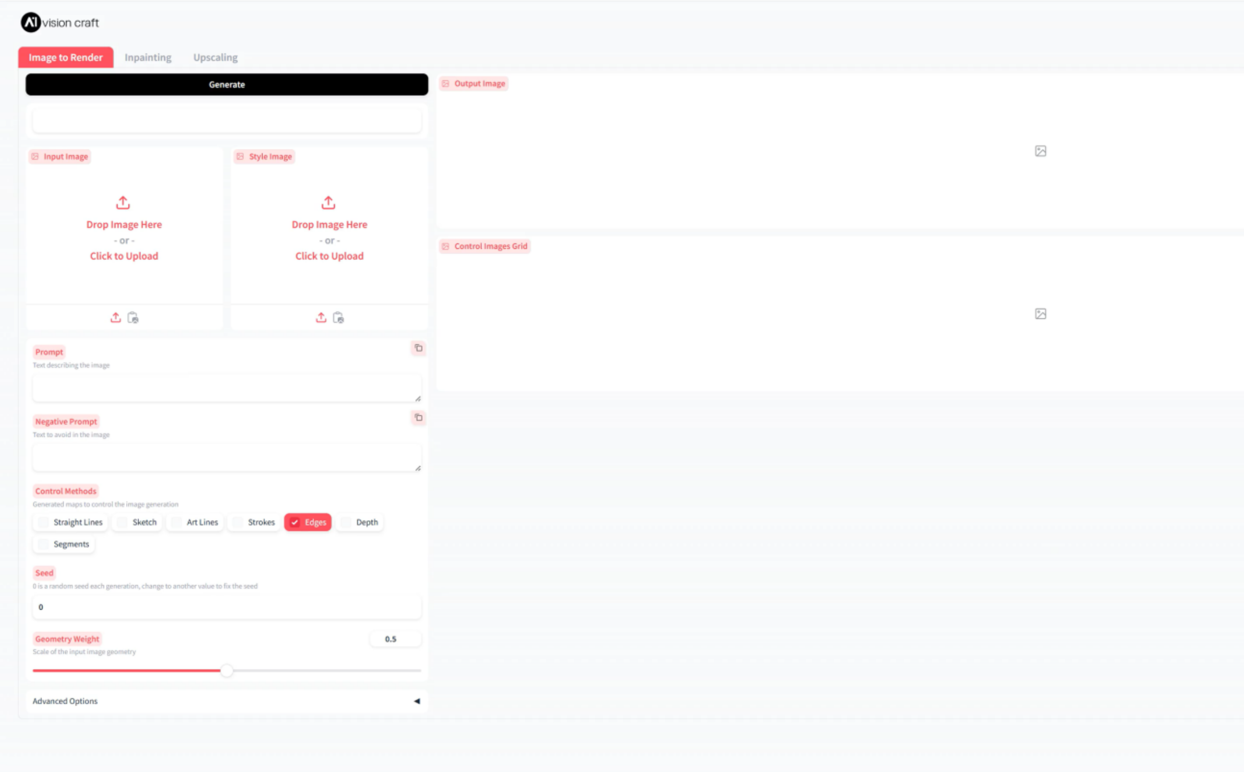
=



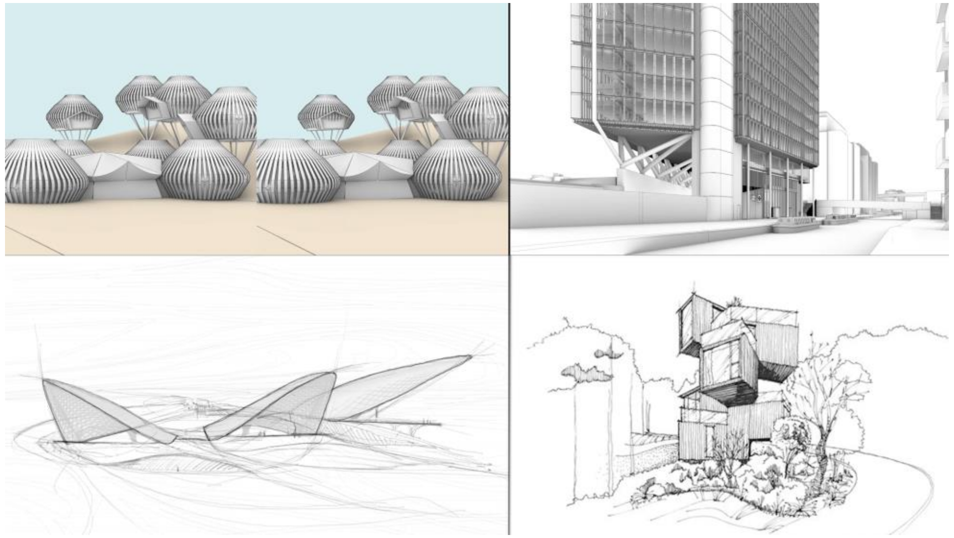
Web based AI Rendering Inhouse tool
Sketch to Render



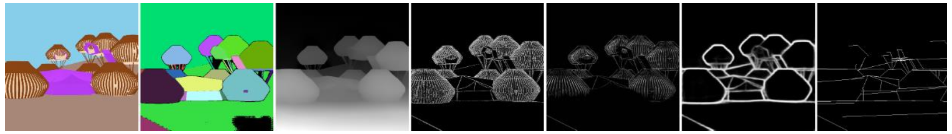
Web based AI Rendering Inhouse tool
Interface – Use case



visioncraft/render.co.uk



Sample Images

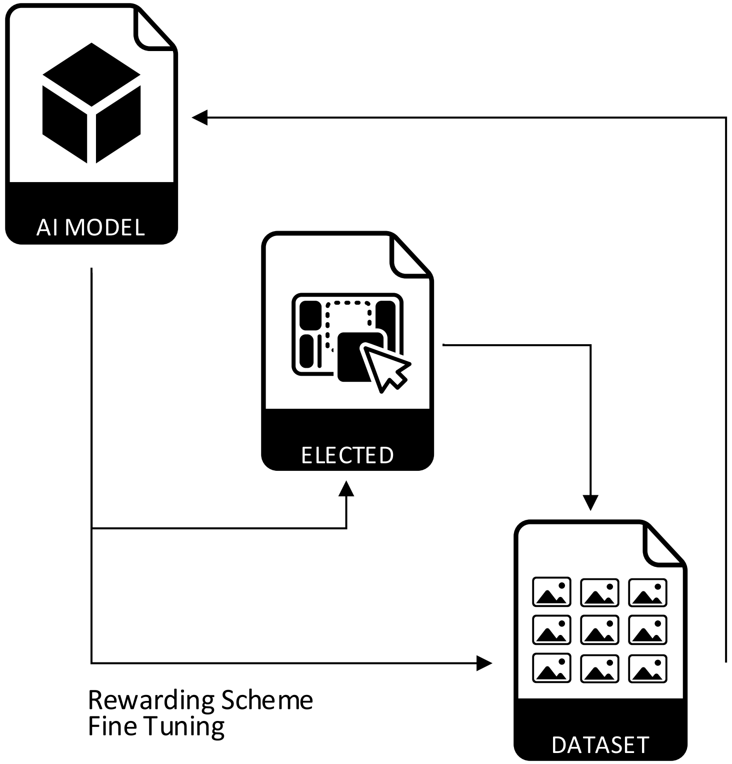


Generative Channels

Web based AI Rendering Inhouse tool
Tool Monitoring

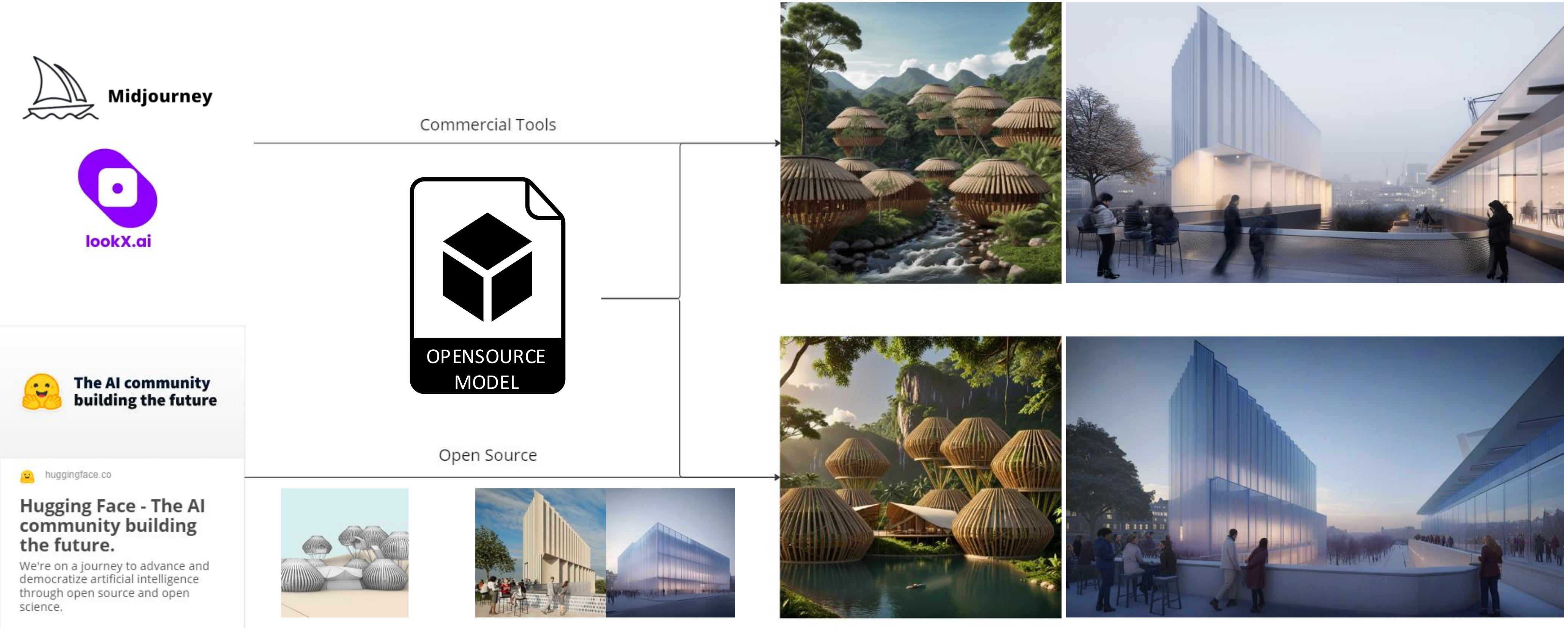
	a stunning view of a cluster of modular pavilions nestled within the lush Brazilian jungle, the	blur,ugly, low res, monochromatic, bad elements, bad images, low pixels	8	4	Seed: 5152658	Edges
	a stunning view of a cluster of modular pavilions nestled within the lush Brazilian jungle, the	blur,ugly, low res, monochromatic, bad elements, bad images, low pixels	8	4	Seed: 8536642	Edges
	a stunning view of a cluster of modular pavilions nestled within the lush Brazilian jungle, the	blur,ugly, low res, monochromatic, bad elements, bad images, low pixels	8	4	Seed: 4982803	Edges
	a stunning view of a cluster of modular pavilions nestled within the lush Brazilian jungle, the	blur,ugly, low res, monochromatic, bad elements, bad images, low pixels	8	4	Seed: 9450913	Edges
	A stunning view of a cluster of modular pavilions nestled within the lush Brazilian jungle at	blur,ugly, low res, monochromatic, bad elements, bad images, low pixels	8	4	Seed: 6276628	Edges
	A stunning view of a cluster of modular pavilions nestled within the lush Brazilian jungle at	blur,ugly, low res, monochromatic, bad elements, bad images, low pixels	8	4	Seed: 8638836	Edges
	A captivating night time view of a cluster of modular pavilions nestled in the misty Brazilian	blur,ugly, low res, monochromatic, bad elements, bad images, low pixels	8	4	Seed: 8652695	Edges
	A captivating night time view of a cluster of modular pavilions nestled in the misty Brazilian	blur,ugly, low res, monochromatic, bad elements, bad images, low pixels	8	3	Seed: 2088714	Edges
	A captivating night time view of a cluster of modular pavilions nestled in the misty Brazilian	blur,ugly, low res, monochromatic, bad elements, bad images, low pixels	8	4	Seed: 6544346	Edges
	A captivating night time view of a cluster of modular pavilions nestled in the misty Brazilian	blur,ugly, low res, monochromatic, bad elements, bad images, low pixels	8	4	Seed: 514236	Straight Lines
	A captivating night time view of a cluster of modular pavilions nestled in the misty Brazilian	blur,ugly, low res, monochromatic, bad elements, bad images, low pixels	8	4	Seed: 5859920	Straight Lines
	A captivating night time view of a cluster of modular pavilions nestled in the misty Brazilian	blur,ugly, low res, monochromatic, bad elements, bad images, low pixels	8	4	Seed: 5407351	Straight Lines
	A captivating night time view of a cluster of modular pavilions nestled in the misty Brazilian	blur,ugly, low res, monochromatic, bad elements, bad images, low pixels	8	4	Seed: 6558849	Straight Lines
	A stunning view of a cluster of modular pavilions nestled within the lush Brazilian jungle at	blur,ugly, low res, monochromatic, bad elements, bad images, low pixels	8	4.1	Seed: 1831188	Straight Lines
	A stunning view of a cluster of modular pavilions nestled within the lush Brazilian jungle at	blur,ugly, low res, monochromatic, bad elements, bad images, low pixels	8	4.1	Seed: 8468774	Straight Lines
	A stunning view of a cluster of modular pavilions nestled within the lush Brazilian jungle at	blur,ugly, low res, monochromatic, bad elements, bad images, low pixels	8	4.1	Seed: 6831563	Straight Lines
	A stunning view of a cluster of modular pavilions nestled within the lush Brazilian jungle at	blur,ugly, low res, monochromatic, bad elements, bad images, low pixels	8	4.1	Seed: 8796608	Straight Lines
	A stunning view of a cluster of modular pavilions nestled within the lush Brazilian jungle at	blur,ugly, low res, monochromatic, bad elements, bad images, low pixels	8	4.1	Seed: 2880361	Sketch

Collected Generated Images
Dataset



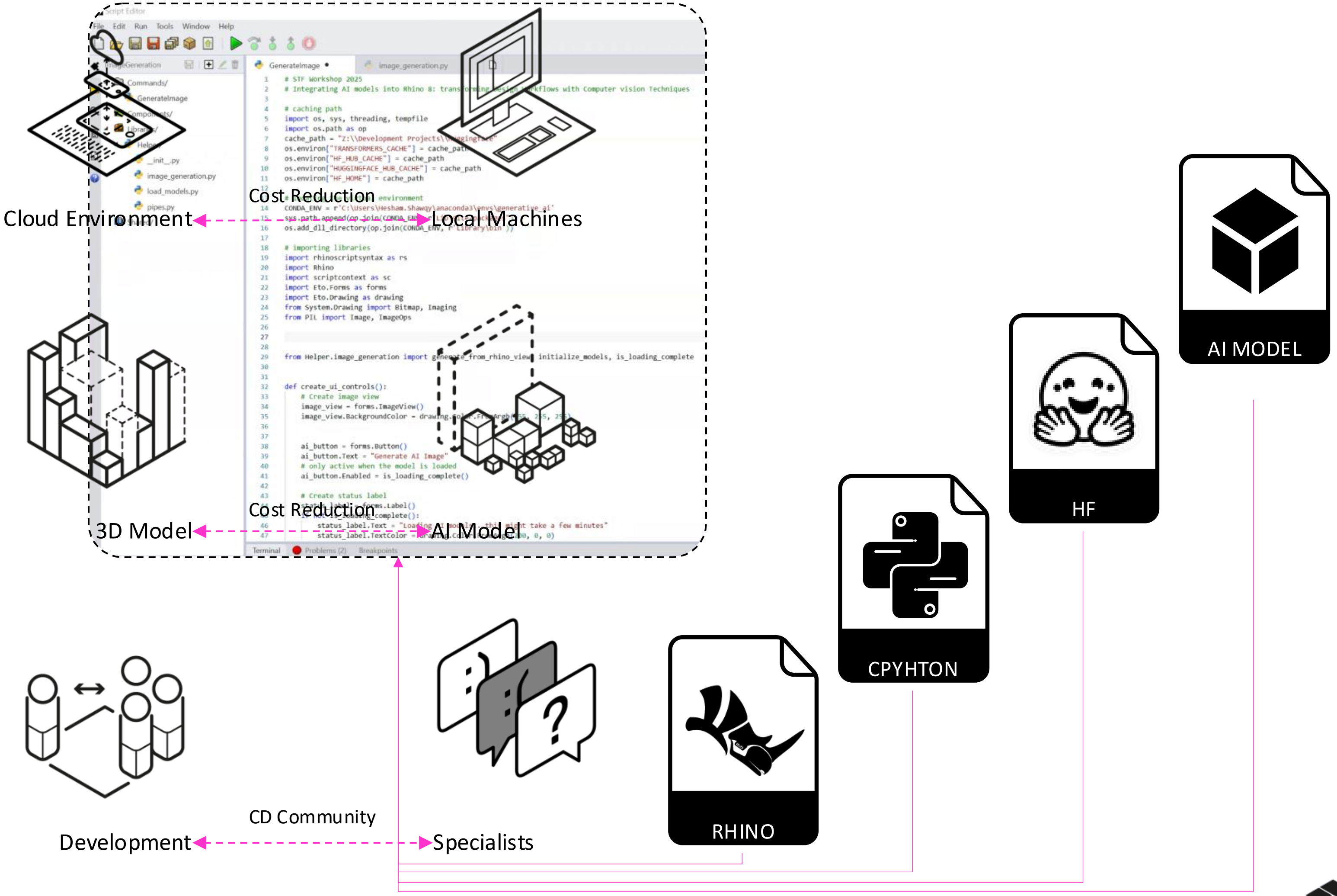
Web based AI Rendering Inhouse tool

Opensource vs commercial tools

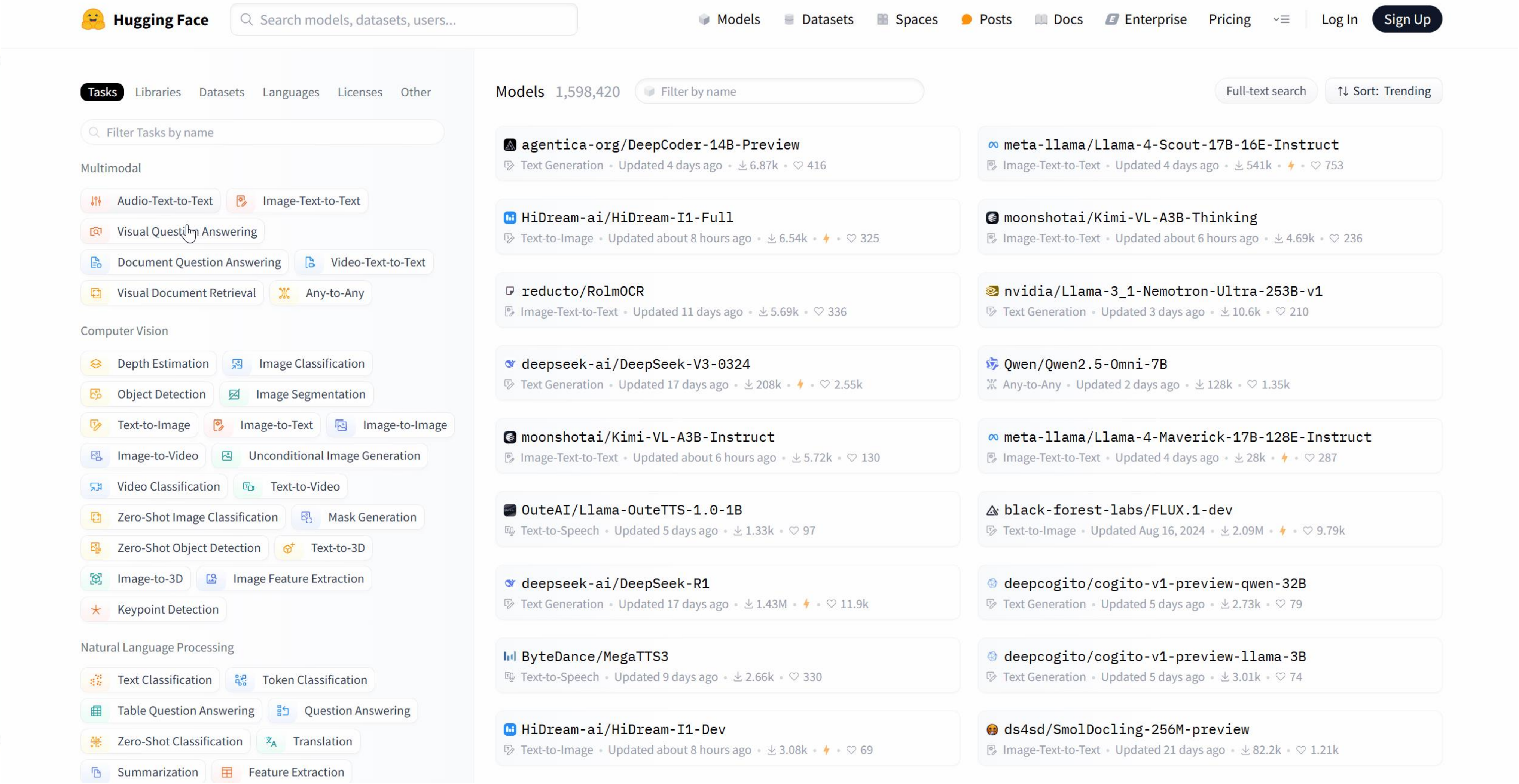


STF WORKSHOP FRAMEWORK

STF Workshop
Why?



STF Workshop
Opensource models



STF Workshop

Rhino 8 Script Editor – Envs - Integration

```python
from diffusers import DiffusionPipeline, StableDiffus
import torch
import gc
from diffusers.utils import load_image
from accelerate.utils import compute_module_sizes

image = load_image("https://huggingface.co/datasets/h

pipe_sd = DiffusionPipeline.from_pretrained("SG161222
pipe_sd.load_ip_adapter("h94/IP-Adapter", subfolder="
pipe_sd.set_ip_adapter_scale(0.6)
pipe_sd.to("cuda")

generator = torch.Generator(device="cpu").manual_seed
out_sd = pipe_sd(
    prompt="bear eats pizza",
    negative_prompt="wrong white balance, dark, sketc
    ip_adapter_image=image,
    num_inference_steps=50,
    generator=generator,
).images[0]
out_sd
```

DiffusionPipeline

diffusers / src / diffusers / pipelines / stable_diffusion / pipeline_stable_diffusion.py

Top

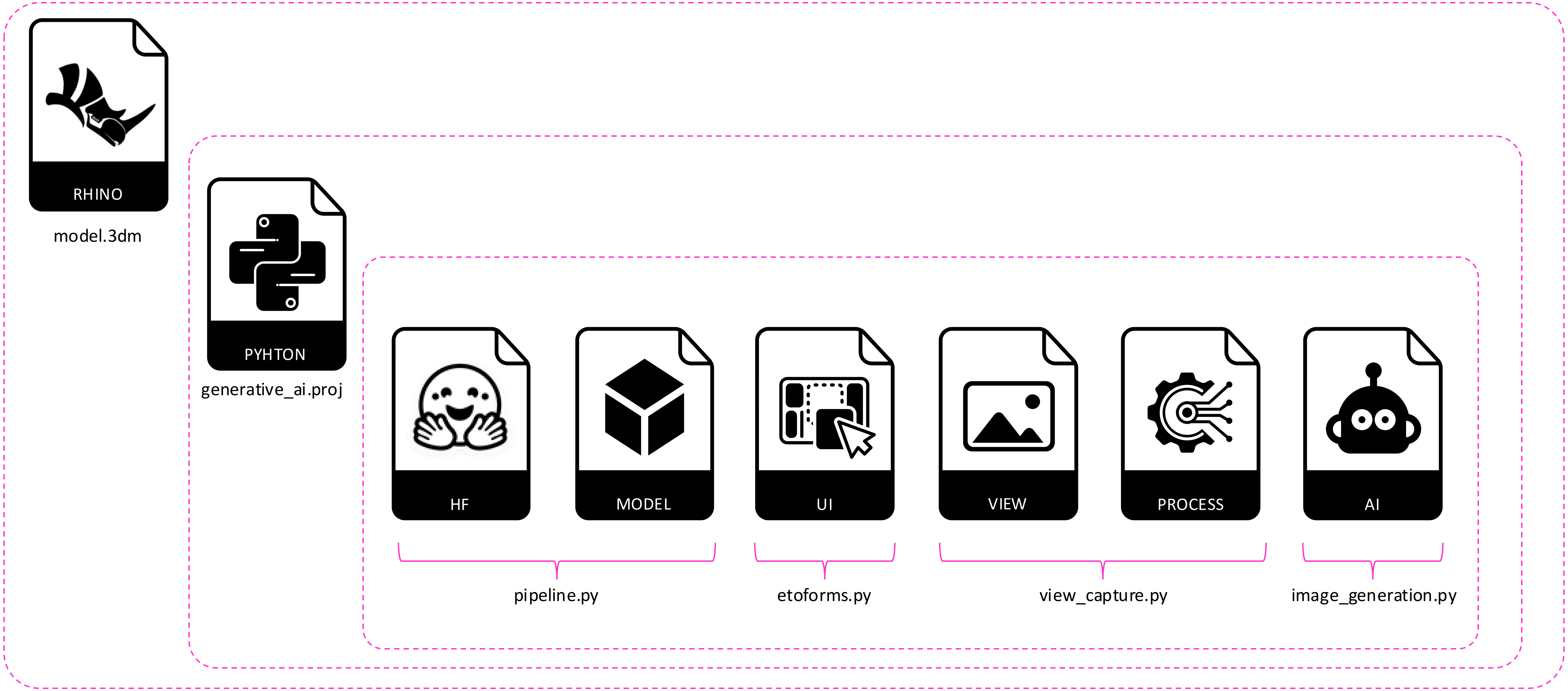
Code Blame 1103 lines (973 loc) · 54.3 KB

Raw

```python
class StableDiffusionPipeline(
    def __call__(
                    prompt_embeds = callback_outputs.pop("prompt_embeds", prompt_embeds)
                    negative_prompt_embeds = callback_outputs.pop("negative_prompt_embeds", negative_prompt_embeds)

                # call the callback, if provided
                if i == len(timesteps) - 1 or ((i + 1) > num_warmup_steps and (i + 1) % self.scheduler.order == 0):
                    progress_bar.update()
                    if callback is not None and i % callback_steps == 0:
                        step_idx = i // getattr(self.scheduler, "order", 1)
                        callback(step_idx, t, latents)

                if XLA_AVAILABLE:
                    xm.mark_step()

        if not output_type == "latent":
            image = self.vae.decode(latents / self.vae.config.scaling_factor, return_dict=False, generator=generator)[
                0
            ]
            image, has_nsfw_concept = self.run_safety_checker(image, device, prompt_embeds.dtype)
        else:
            image = latents
            has_nsfw_concept = None

        if has_nsfw_concept is None:
            do_denormalize = [True] * image.shape[0]
        else:
            do_denormalize = [not has_nsfw for has_nsfw in has_nsfw_concept]
        image = self.image_processor.postprocess(image, output_type=output_type, do_denormalize=do_denormalize)

        # Offload all models
        self.maybe_free_model_hooks()

        if not return_dict:
            return (image, has_nsfw_concept)

```

Diffusers

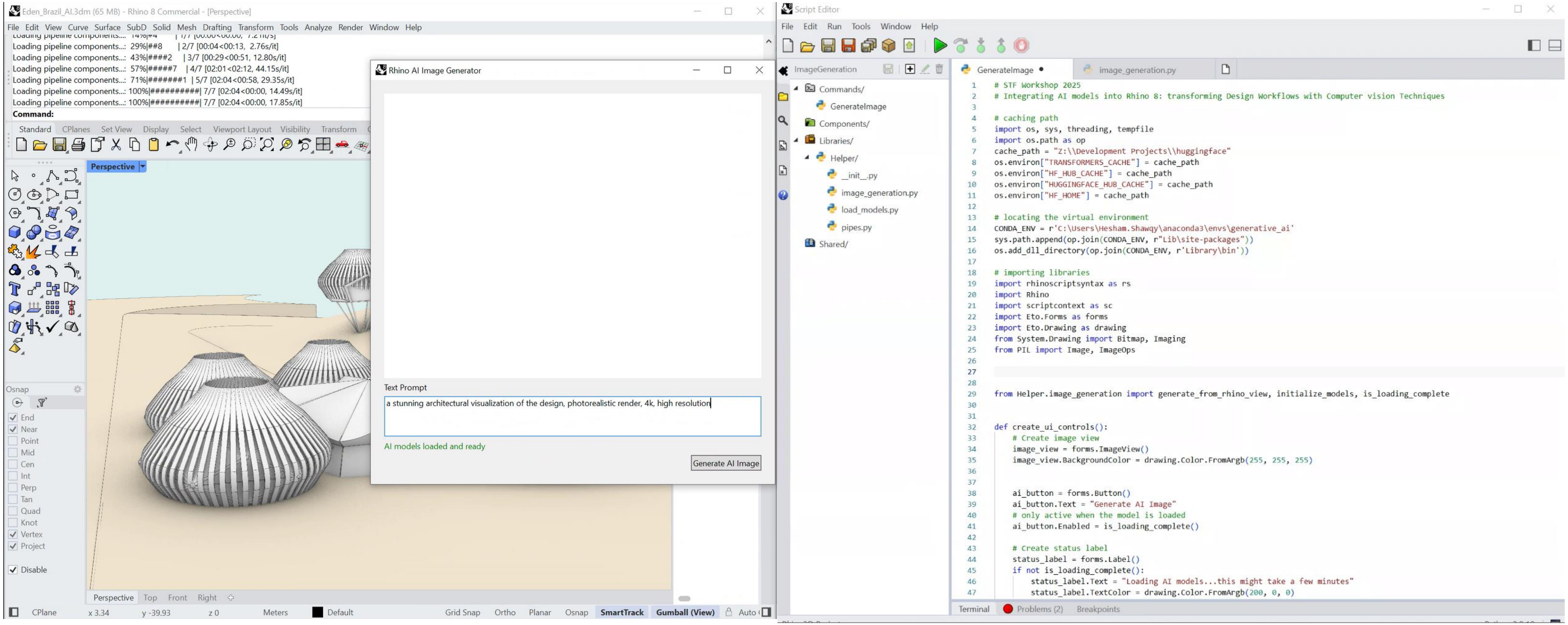
STF Workshop
Rhino 8 Script Editor – Envs - Integration



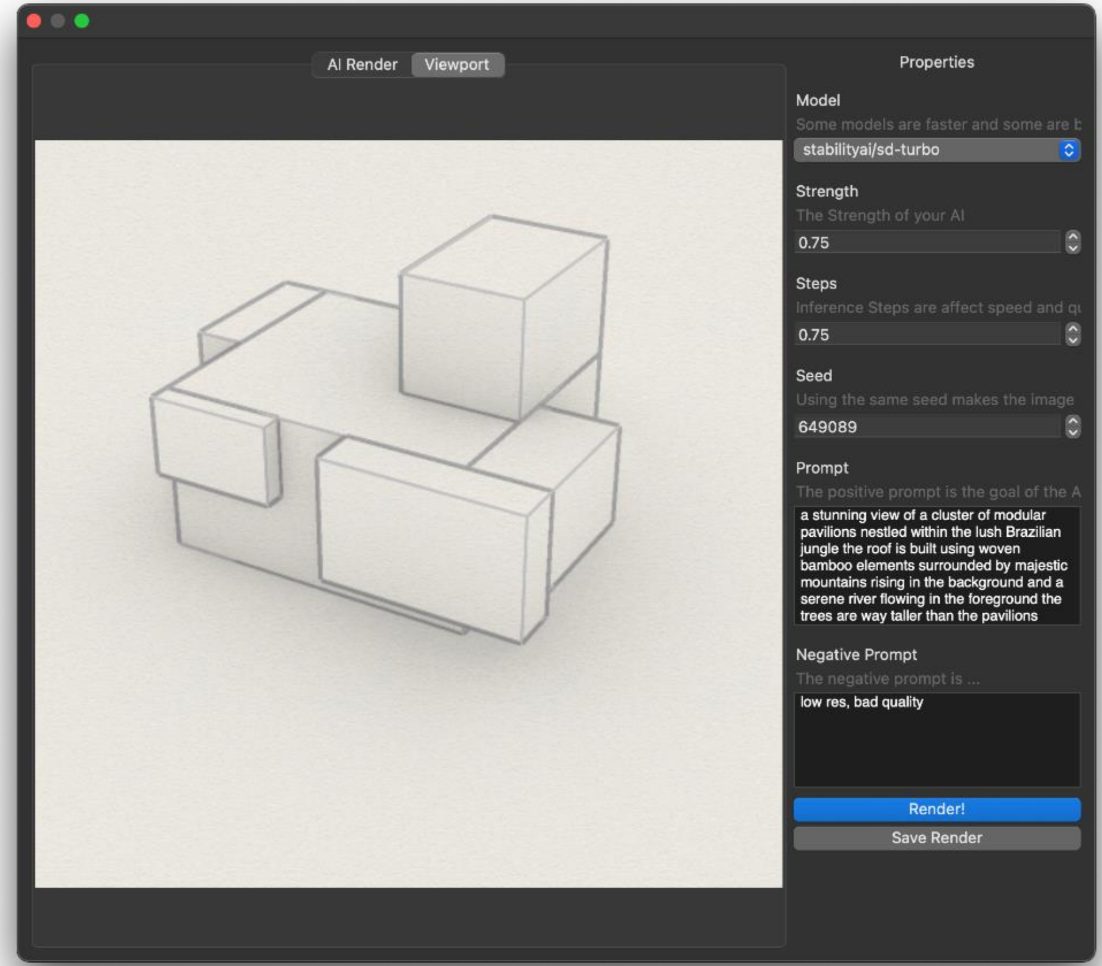
STF WORKSHOP
SCRIPT EDITOR

STF Workshop

User Interface – Eto.Forms



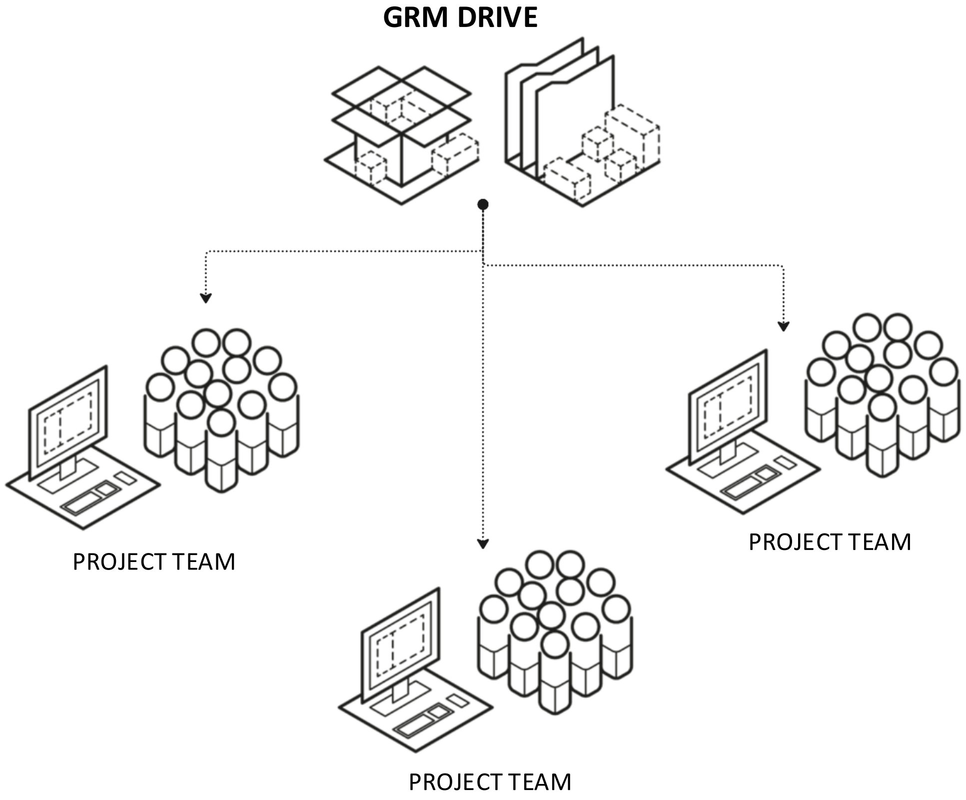
STF Workshop
User Interface – Eto.Forms

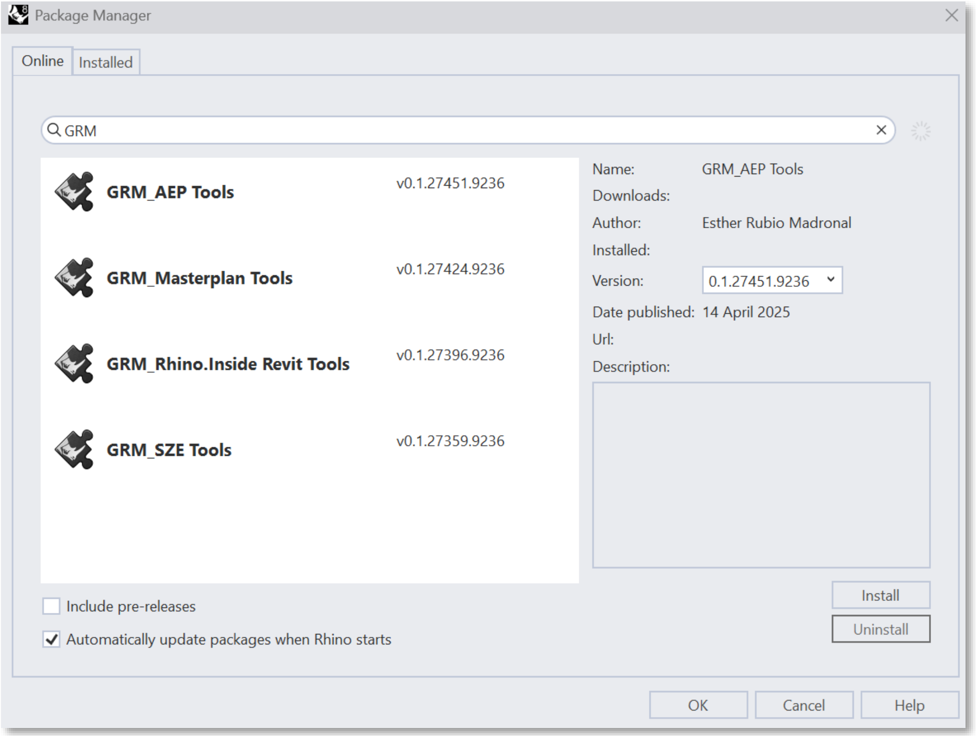


- Choose model to work with
- Change parameters
- Control seed
- Prompt & Neg.Prompt

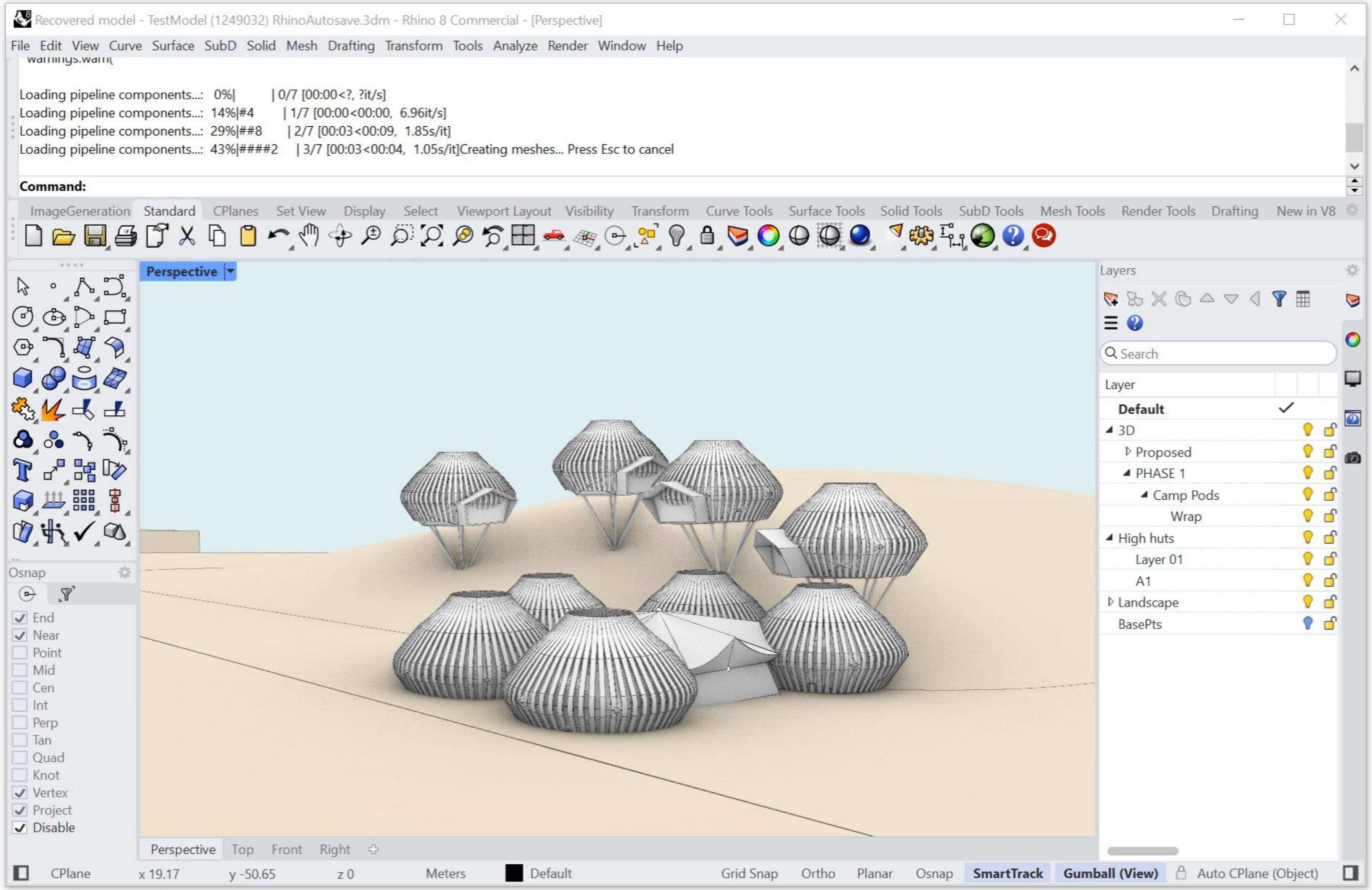
By Callum Sykes – thanks !

STF Workshop
Project Plugins

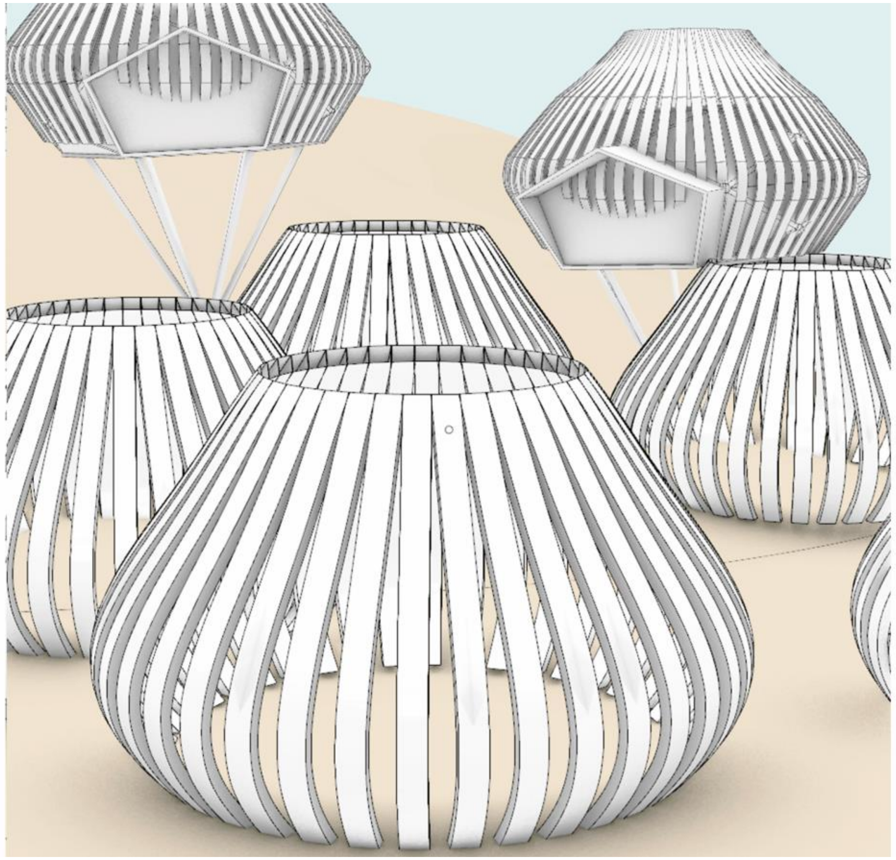




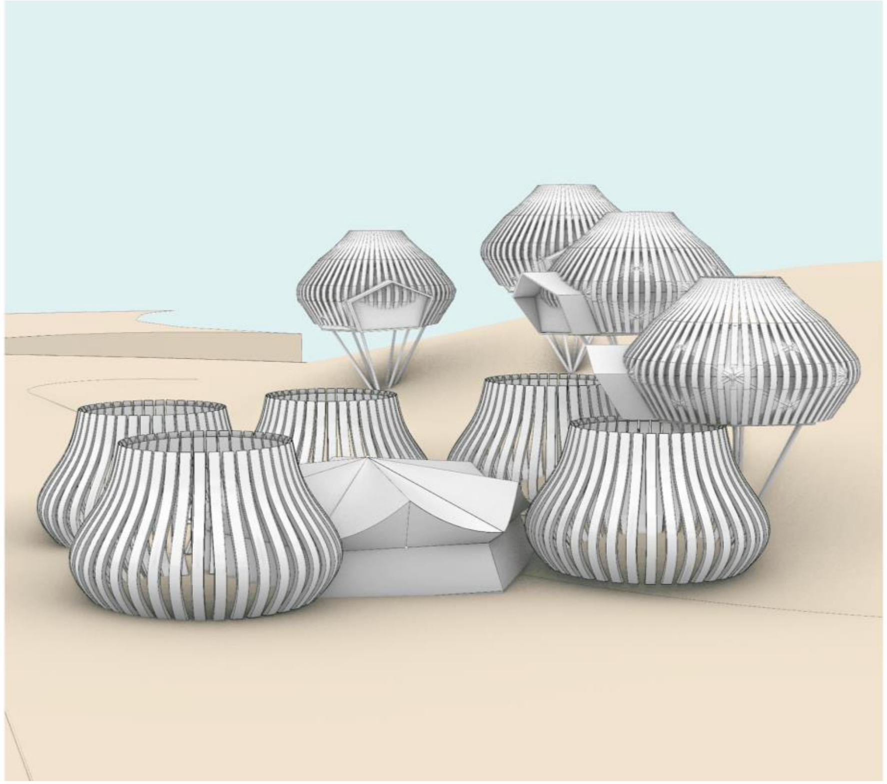
STF Workshop
Projects-Plugins

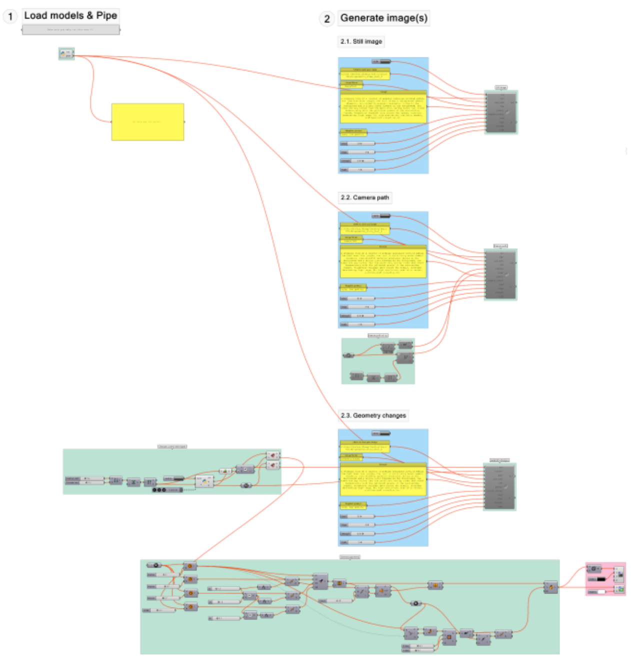


STF Workshop
Grasshopper

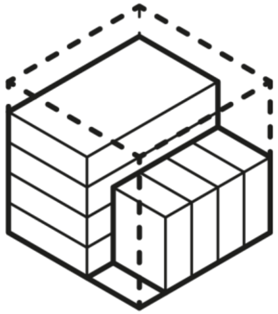


STF Workshop
Grasshopper



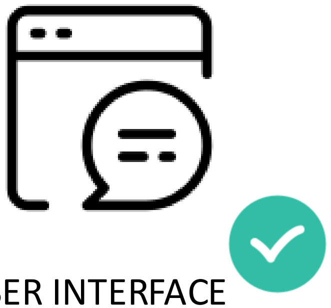


STF Workshop
Challenges

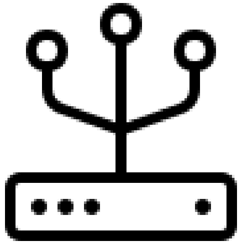


AI MODELS SIZE

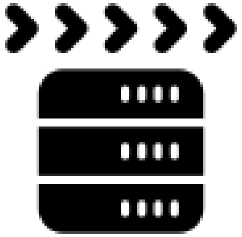
SHARED ENVIRONMENT



USER INTERFACE



HARDWARE



CONSISTENCY

MEMORY MANAGEMENT

STF Workshop
Challenges

STF WORKSHOP
BEYOND RENDERING

EXTENDED WORKFLOWS

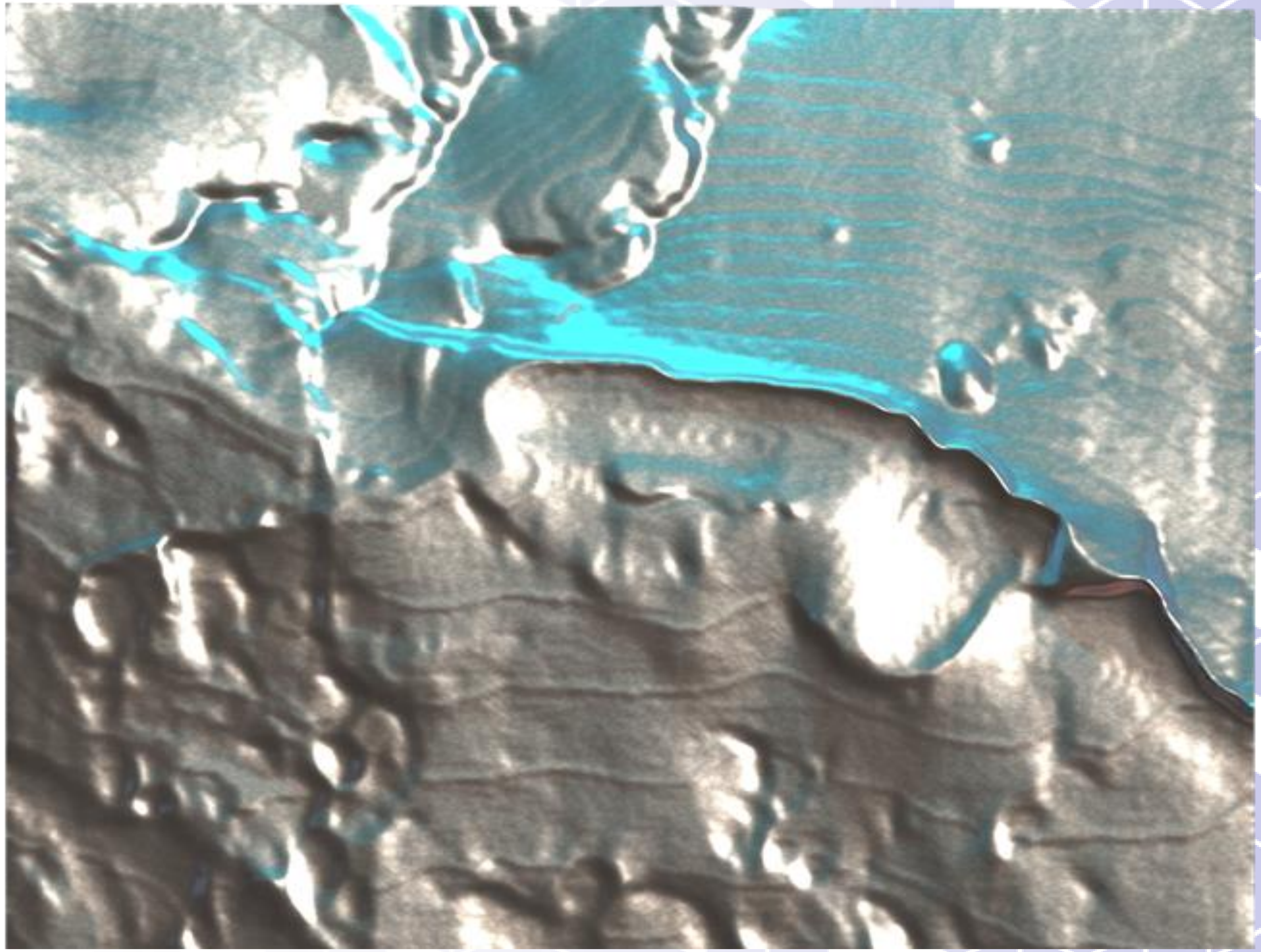
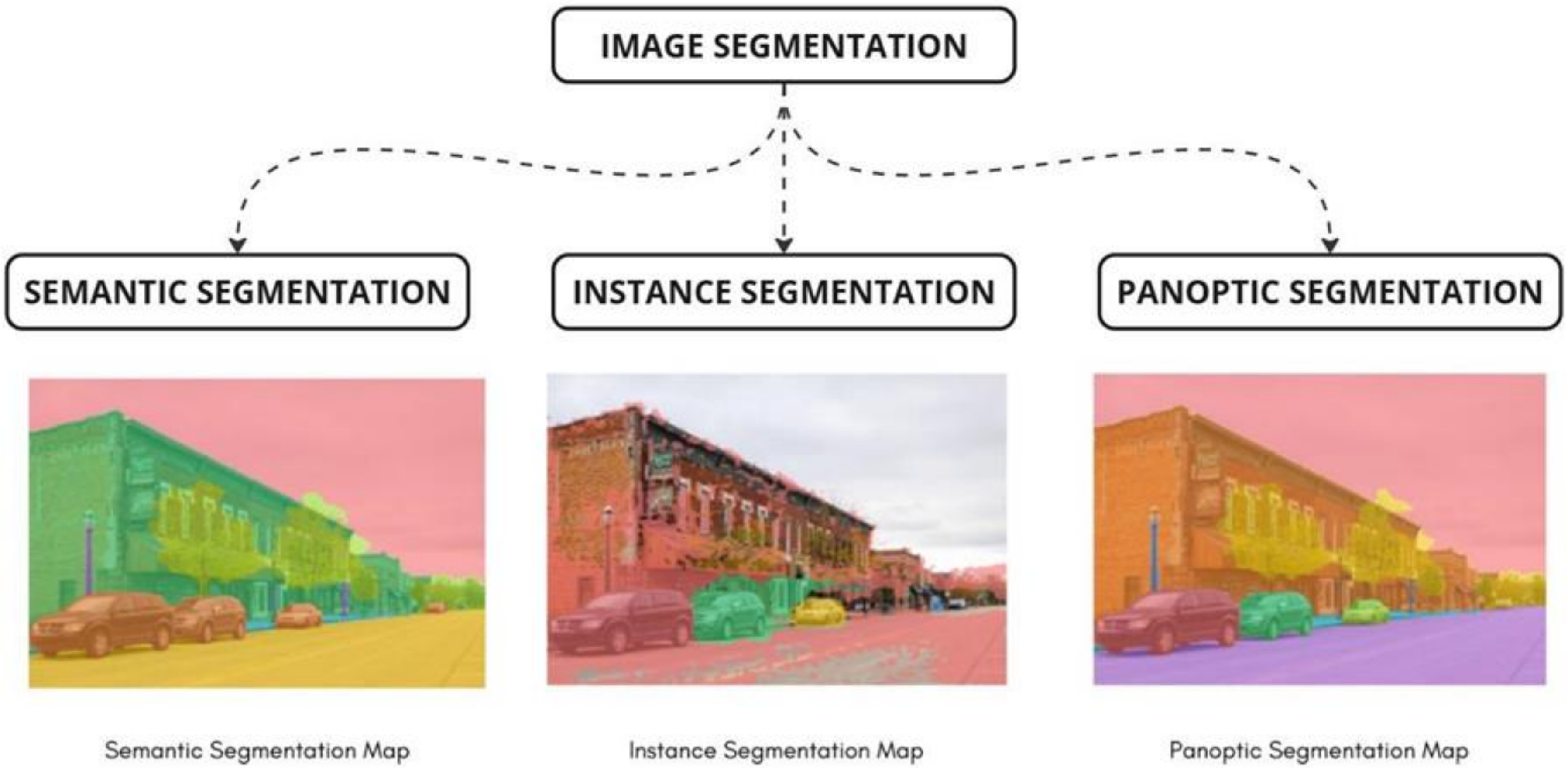


Image Segmentation



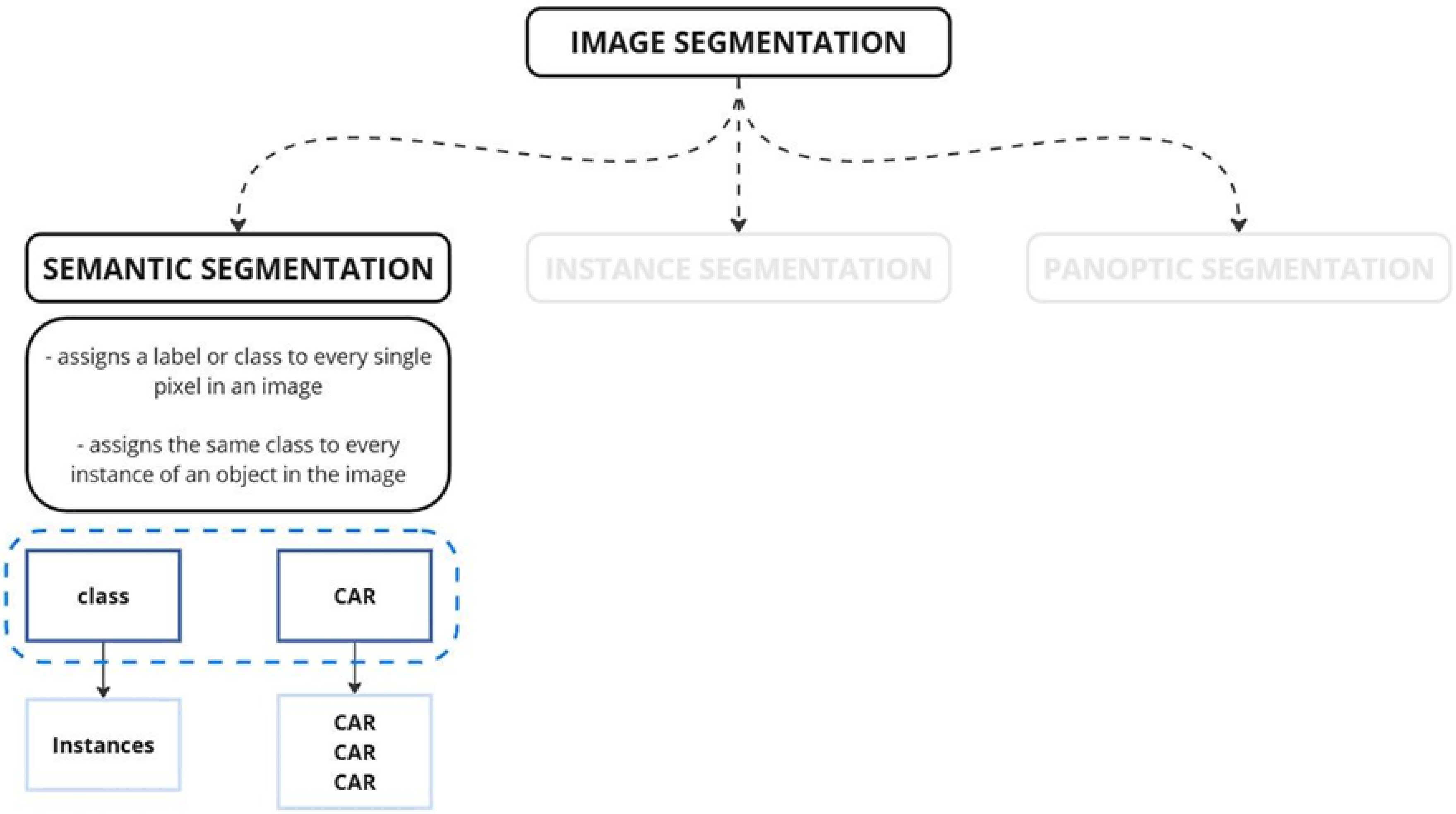


IMAGE SEGMENTATION
SEMANTIC SEGMENTATION
INSTANCE SEGMENTATION
PANOPTIC SEGMENTATION
- assigns a label or class to every single pixel in an image
- assigns the same class to every instance of an object in the image
class
CAR
Instances
CAR
CAR
CAR

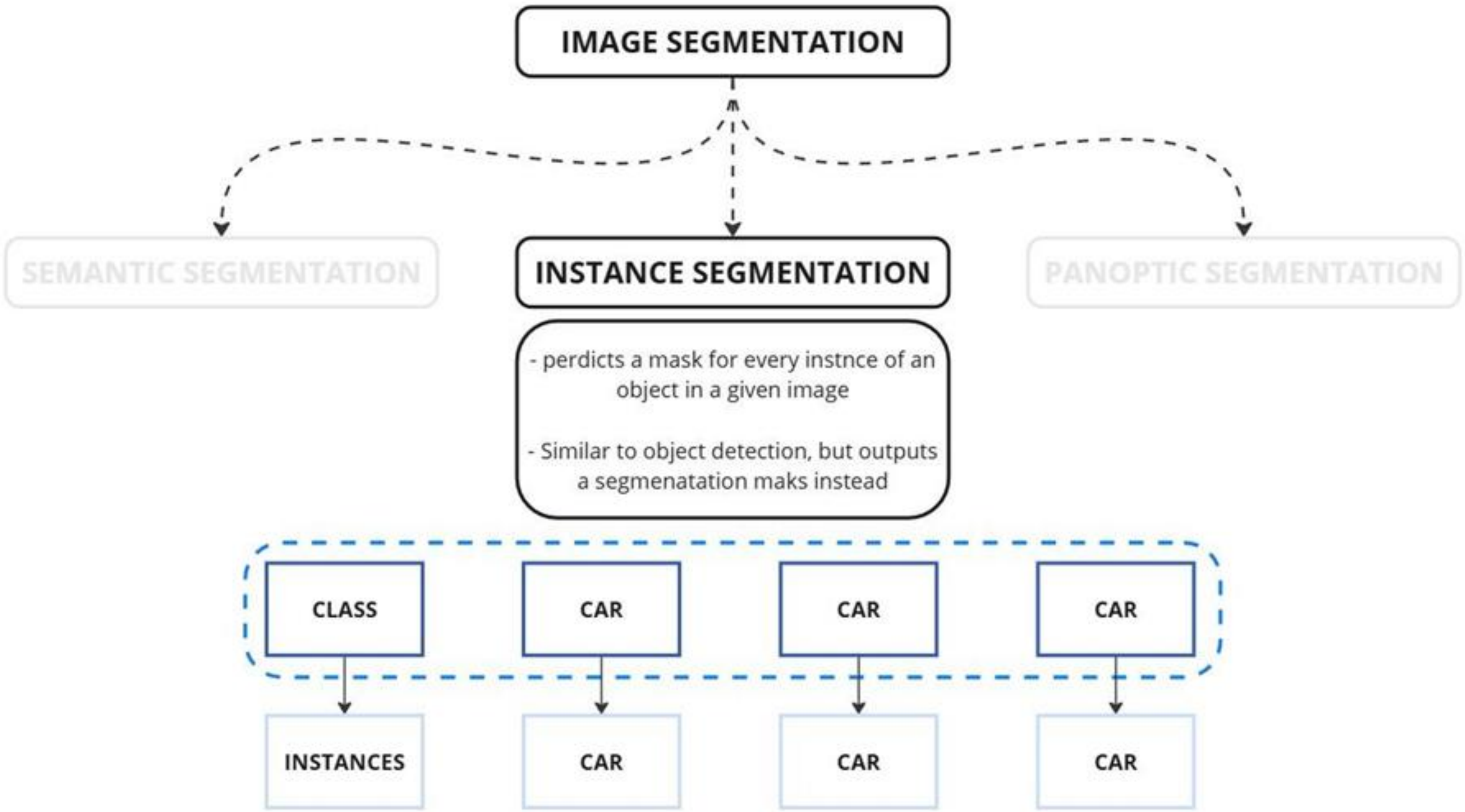


IMAGE SEGMENTATION
SEMANTIC SEGMENTATION
INSTANCE SEGMENTATION
PANOPTIC SEGMENTATION
- perdicts a mask for every instnce of an object in a given image
- Similar to object detection, but outputs a segmenatation maks instead
CLASS
CAR
CAR
CAR
INSTANCES
CAR
CAR
CAR

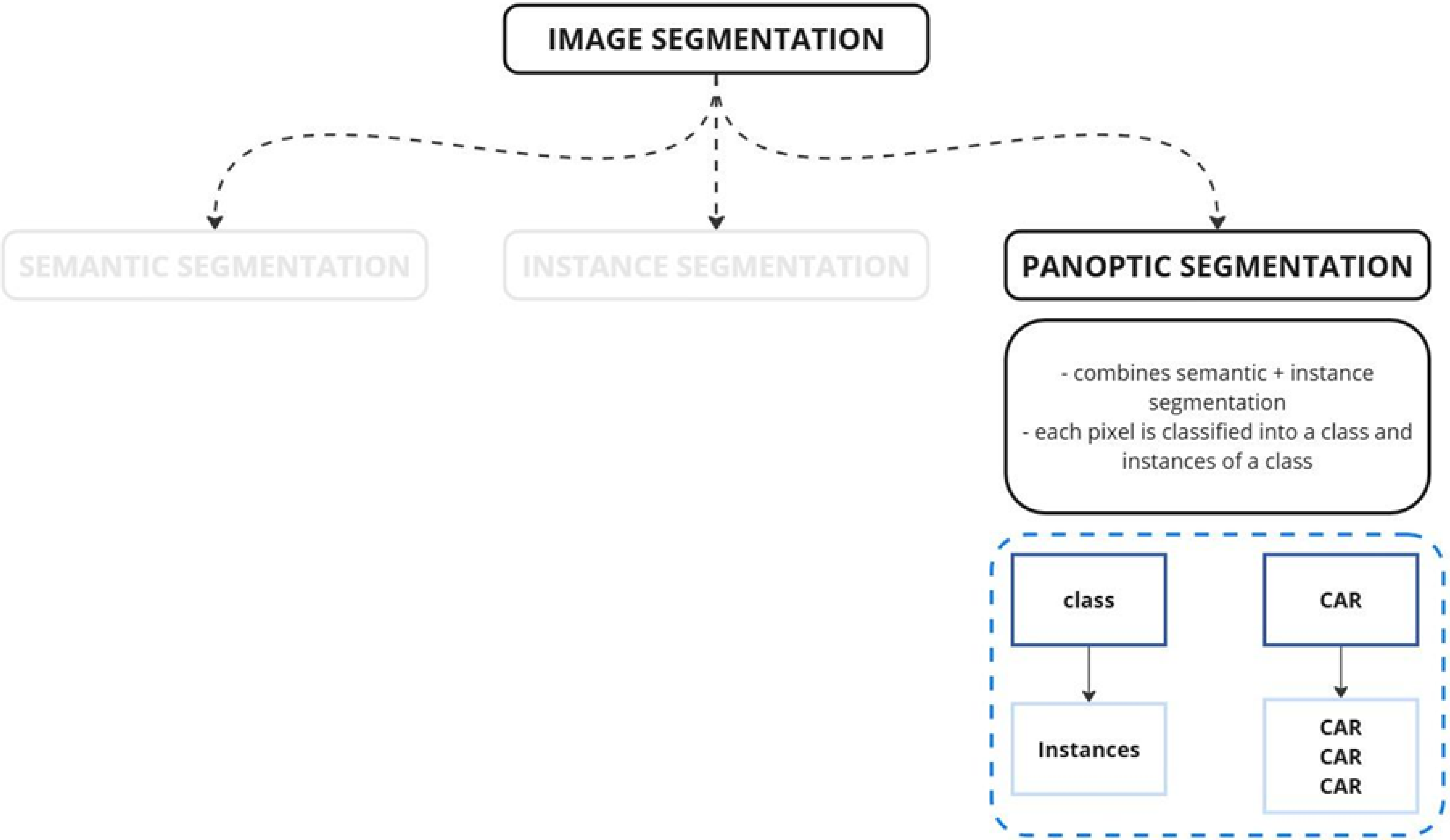


IMAGE SEGMENTATION
SEMANTIC SEGMENTATION
INSTANCE SEGMENTATION
PANOPTIC SEGMENTATION
- combines semantic + instance segmentation
- each pixel is classified into a class and instances of a class
class
Instances
CAR
CAR
CAR
CAR

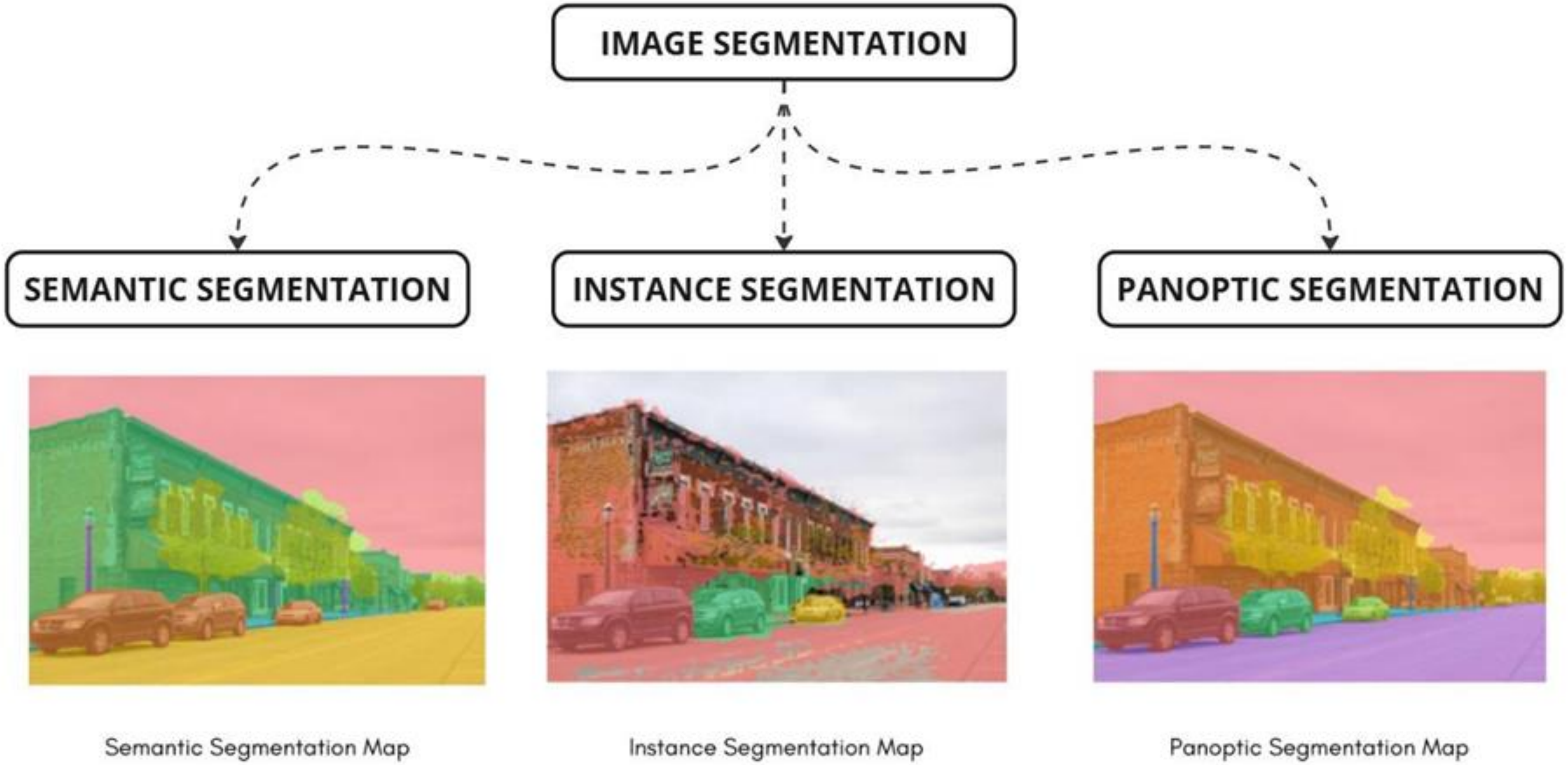
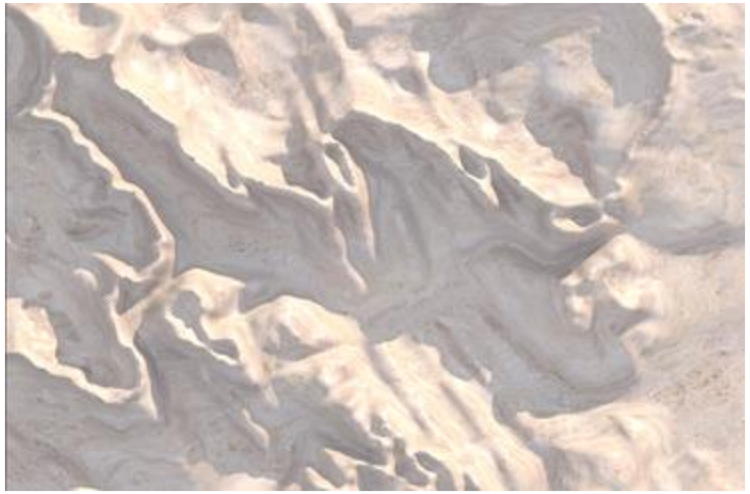
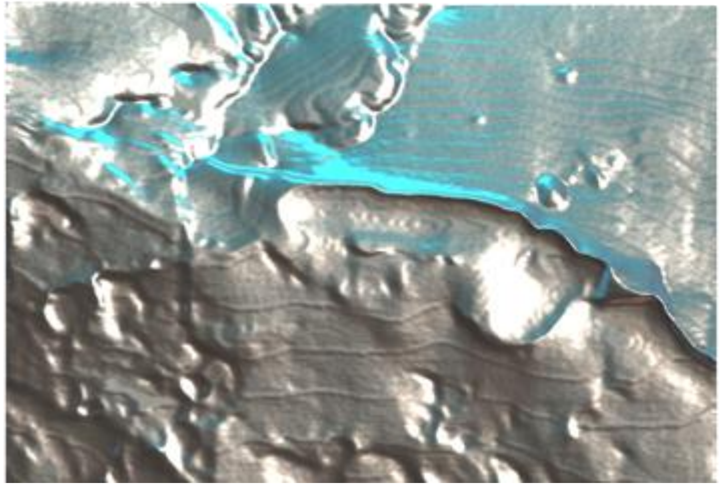


IMAGE SEGMENTATION
SEMANTIC SEGMENTATION
INSTANCE SEGMENTATION
PANOPTIC SEGMENTATION
Semantic Segmentation Map
Instance Segmentation Map
Panoptic Segmentation Map

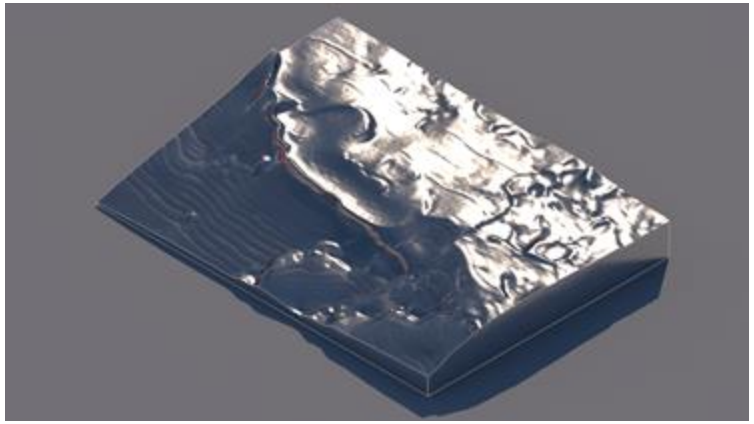
Depth Estimation





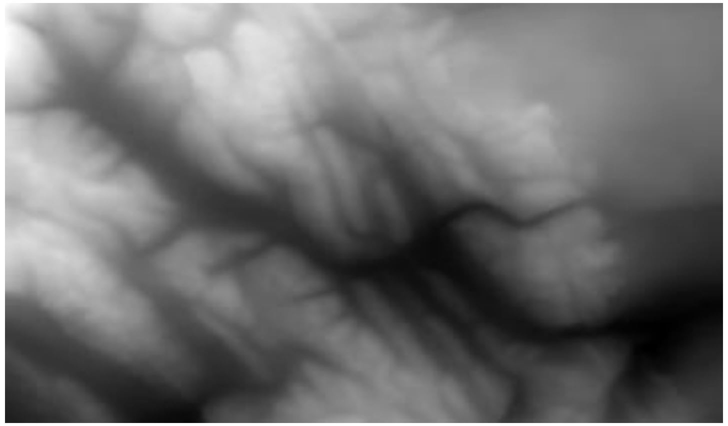
Input
image

Ai
generated
Depth
Map

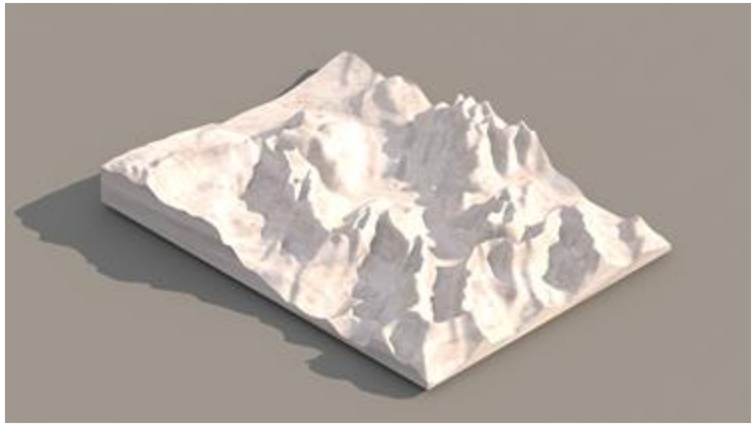


Resulting form generated in
grasshopper

Input image



Ai generated Depth Map



Resulting form generated in grasshopper

Object Detection

confidential

STF WORKSHOP
BEYOND RENDERING

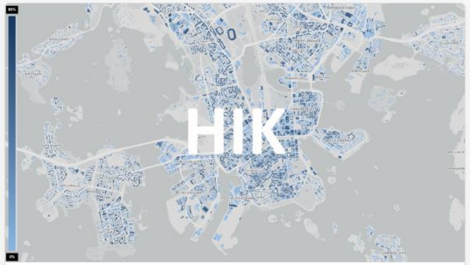
Beyond AI rendering
Urban Mining, 2021

The material library map of Barcelona of **59743** buildings data points
coordinates **[2.20, 41.40]**

Explore

The material library map of Delhi of **24169** buildings data points
coordinates **[77.18,28.58]**

Explore



The material library map of Helsinki of **62408** buildings data points
coordinates **[24.97,60.20]**

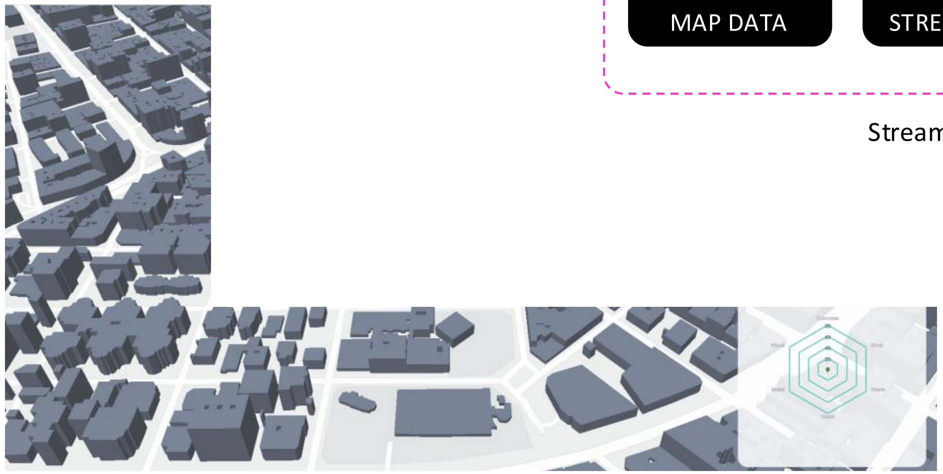
Explore

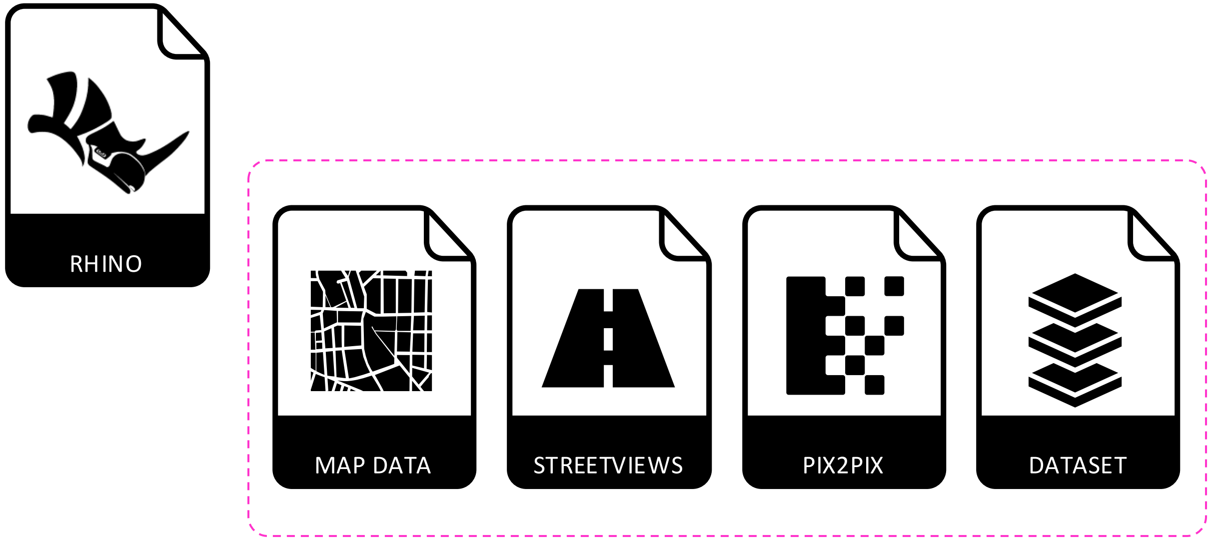
The material library map of Singapore of **110768** buildings data points
coordinates **[103.82,1.32]**

Explore

Enhanced Databases on City’s Building Material Stock. An Urban Mining Method Based on Machine Learning for Enabling Building’ s Materials Reuse Strategies.
Areti Markopoulou, Oana Taut, Hesham Shawqy.
Chapter of Design for Rethinking Resources Proceedings of the UIA World Congress of Architects Copenhagen 2023

Beyond AI rendering
Urban Mining, 2021



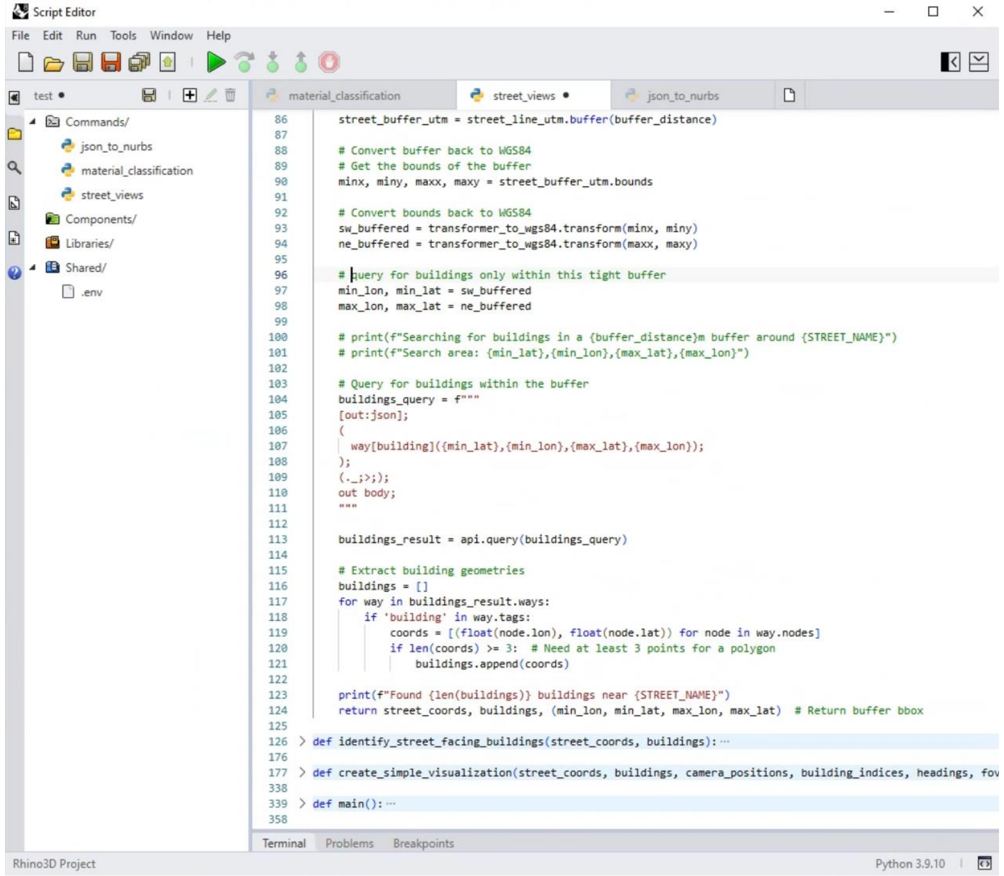


Streamlined process inside Rhino3d

Urban Imagery Data

Query buildings’ features

OVERPASS API

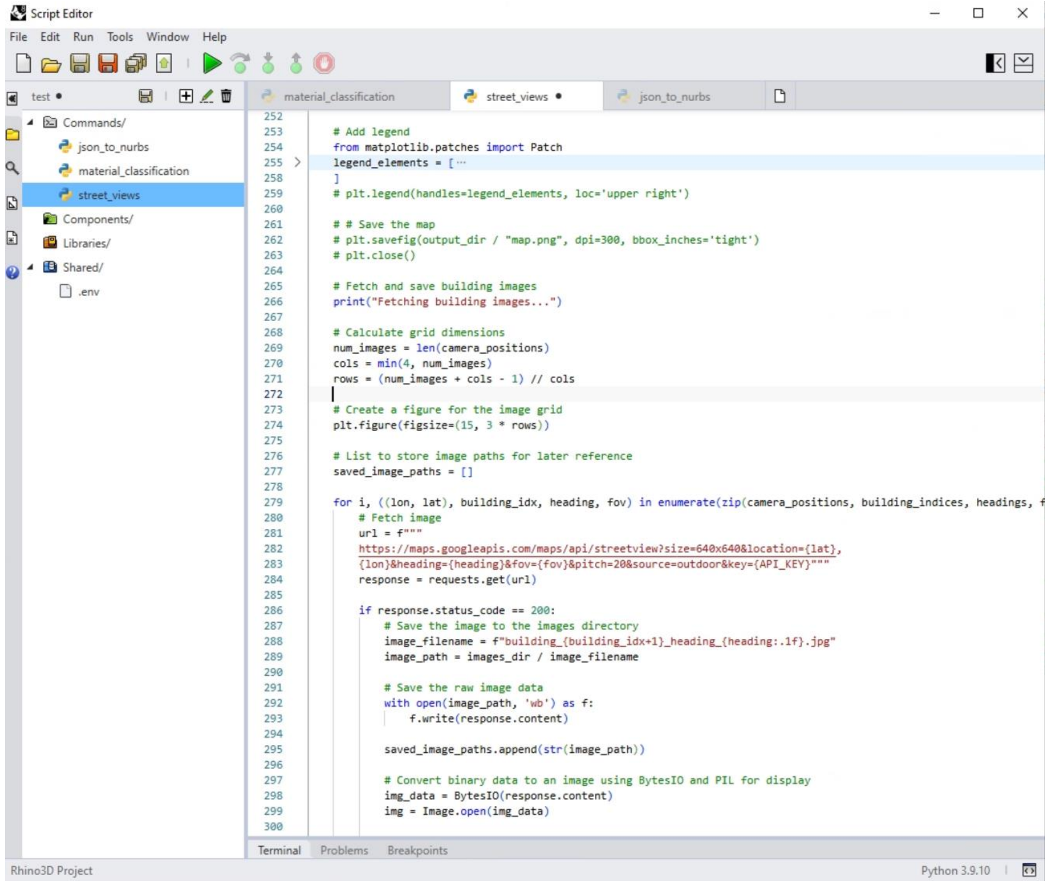



```python
    street_buffer_utm = street_line_utm.buffer(buffer_distance)

    # Convert buffer back to WGS84
    # Get the bounds of the buffer
    minx, miny, maxx, maxy = street_buffer_utm.bounds

    # Convert bounds back to WGS84
    sw_buffered = transformer_to_wgs84.transform(minx, miny)
    ne_buffered = transformer_to_wgs84.transform(maxx, maxy)

    # query for buildings only within this tight buffer
    min_lon, min_lat = sw_buffered
    max_lon, max_lat = ne_buffered

    # print(f"Searching for buildings in a {buffer_distance}m buffer around {STREET_NAME}")
    # print(f"Search area: {min_lat},{min_lon},{max_lat},{max_lon}")

    # Query for buildings within the buffer
    buildings_query = f"""
    [out:json];
    (
      way[building]({min_lat},{min_lon},{max_lat},{max_lon});
    );
    (._;>;);
    out body;
    """

    buildings_result = api.query(buildings_query)

    # Extract building geometries
    buildings = []
    for way in buildings_result.ways:
        if 'building' in way.tags:
            coords = [(float(node.lon), float(node.lat)) for node in way.nodes]
            if len(coords) >= 3:  # Need at least 3 points for a polygon
                buildings.append(coords)

    print(f"Found {len(buildings)} buildings near {STREET_NAME}")
    return street_coords, buildings, (min_lon, min_lat, max_lon, max_lat)  # Return buffer bbox

def identify_street_facing_buildings(street_coords, buildings): ...

def create_simple_visualization(street_coords, buildings, camera_positions, building_indices, headings, fov

def main(): ...
```

Urban Imagery Data

Query buildings’ streetviews

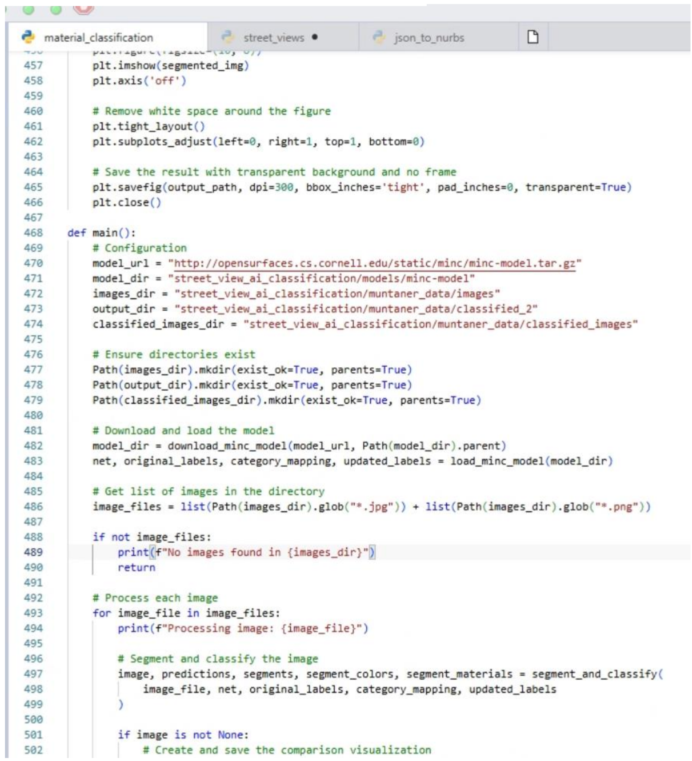
GOOGLE STREETVIEWS API



Urban Imagery Data
Image Material Classification

opensurfaces.cs.cornell.edu/publications/minc/models

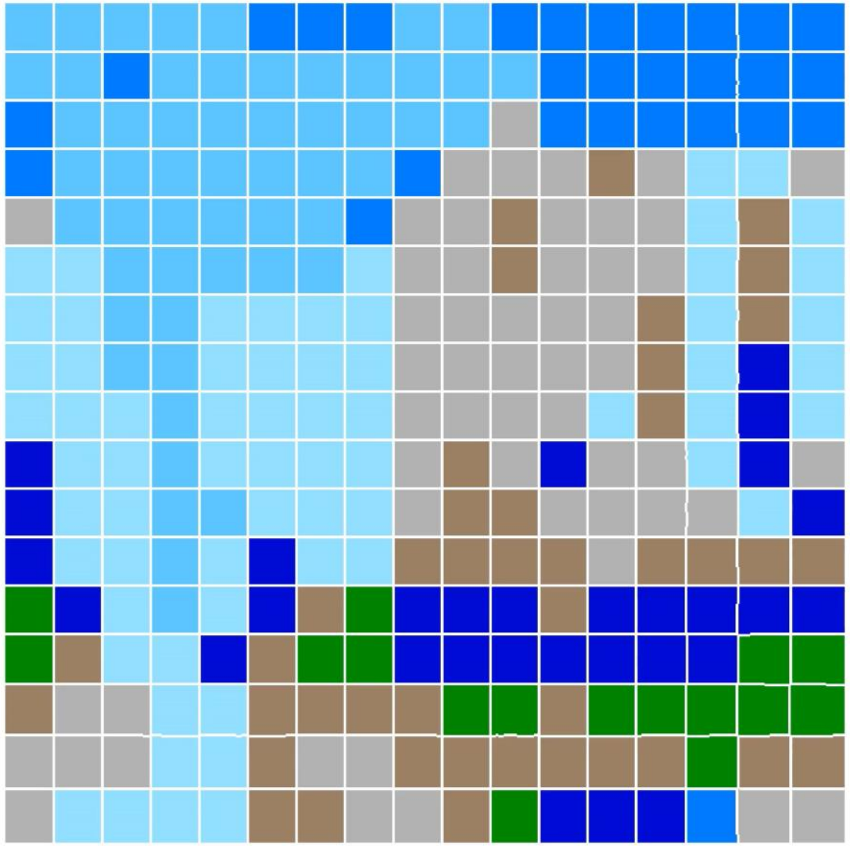
Hugging Face MINC-dataset




```python
    plt.imshow(segmented_img)
    plt.axis('off')

    # Remove white space around the figure
    plt.tight_layout()
    plt.subplots_adjust(left=0, right=1, top=1, bottom=0)

    # Save the result with transparent background and no frame
    plt.savefig(output_path, dpi=300, bbox_inches='tight', pad_inches=0, transparent=True)
    plt.close()

def main():
    # Configuration
    model_url = "http://opensurfaces.cs.cornell.edu/static/minc/minc-model.tar.gz"
    model_dir = "street_view_ai_classification/models/minc-model"
    images_dir = "street_view_ai_classification/muntaner_data/images"
    output_dir = "street_view_ai_classification/muntaner_data/classified_2"
    classified_images_dir = "street_view_ai_classification/muntaner_data/classified_images"

    # Ensure directories exist
    Path(images_dir).mkdir(exist_ok=True, parents=True)
    Path(output_dir).mkdir(exist_ok=True, parents=True)
    Path(classified_images_dir).mkdir(exist_ok=True, parents=True)

    # Download and load the model
    model_dir = download_minc_model(model_url, Path(model_dir).parent)
    net, original_labels, category_mapping, updated_labels = load_minc_model(model_dir)

    # Get list of images in the directory
    image_files = list(Path(images_dir).glob("*.jpg")) + list(Path(images_dir).glob("*.png"))

    if not image_files:
        print(f"No images found in {images_dir}")
        return

    # Process each image
    for image_file in image_files:
        print(f"Processing image: {image_file}")

        # Segment and classify the image
        image, predictions, segments, segment_colors, segment_materials = segment_and_classify(
            image_file, net, original_labels, category_mapping, updated_labels
        )

        if image is not None:
            # Create and save the comparison visualization
```

Original Image

Material Classification



Material Recognition in the Wild with the Materials in Context Database. Sean Bell, Paul Upchurch, Noah Snavely, Kavita Bala Cornell University

Urban Imagery Data

Image Semantic Classification

nvidia/segformer-b1-finetuned-cityscapes-1024-1024

Hugging Face MINC-dataset

material_classification | street_views | json_to_nurbs

```python
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```
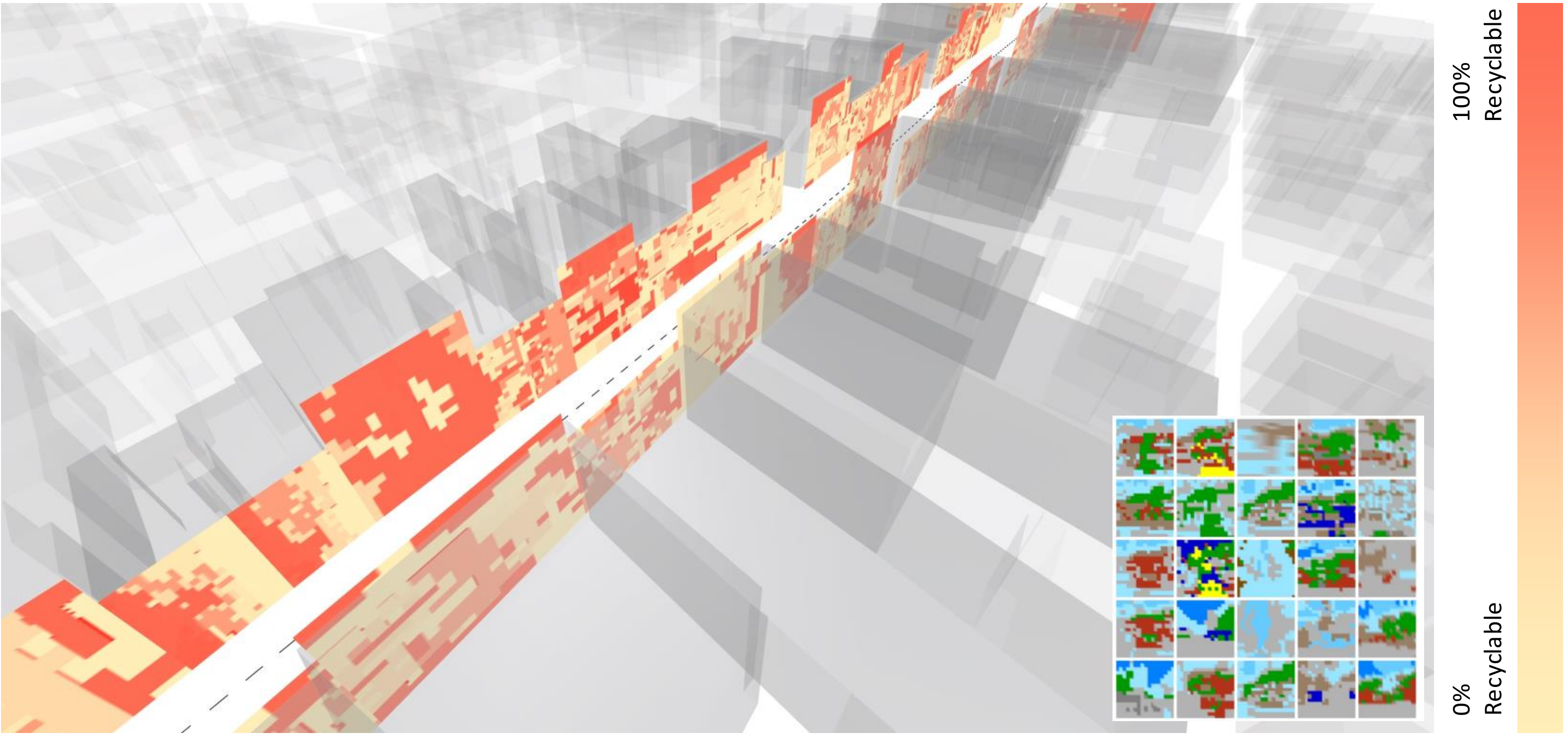

- bicycle
- building
- car
- fence
- motorcycle
- person
- pole
- rider
- road
- sidewalk
- sky
- traffic sign
- vegetation

Urban Imagery Data
Image Object Detection

Urban Imagery Data
Heatmaps

image_name	brick	glass	metal	flective_gla	sky	stone	vegetation	water	wood
building_100_heading_38.7.jpg	0.0	0.0	32.5	26.0	20.8	0.0	10.4	0.0	0.0
building_111_heading_222.8.jpg	23.2	6.2	46.7	15.6	0.0	0.0	8.3	0.0	0.0
building_112_heading_222.8.jpg	24.2	7.6	32.9	25.3	0.0	10.0	0.0	0.0	0.0
building_113_heading_222.8.jpg	8.3	3.8	11.8	26.6	0.0	28.7	20.8	0.0	0.0
building_115_heading_222.8.jpg	11.8	7.3	27.7	19.7	0.0	17.6	15.9	0.0	0.0
building_117_heading_222.7.jpg	25.6	0.0	38.4	13.5	0.0	11.4	11.1	0.0	0.0
building_118_heading_220.1.jpg	0.0	0.0	43.6	15.9	20.1	0.0	15.2	5.2	0.0
building_140_heading_221.0.jpg	9.0	0.0	23.2	7.3	0.0	0.0	23.9	22.5	0.0
building_141_heading_221.0.jpg	0.0	0.0	37.7	23.9	0.0	0.0	38.4	0.0	0.0
building_142_heading_221.0.jpg	19.0	0.0	26.0	18.0	0.0	11.1	12.8	0.0	0.0
building_143_heading_221.0.jpg	0.0	4.5	28.4	32.2	0.0	15.6	19.4	0.0	0.0
building_145_heading_221.0.jpg	0.0	26.3	54.0	19.7	0.0	0.0	0.0	0.0	0.0
building_146_heading_221.0.jpg	0.0	4.5	11.1	73.0	0.0	0.0	2.4	0.0	9.0
building_147_heading_221.0.jpg	0.0	4.5	28.4	32.2	0.0	15.6	19.4	0.0	0.0
building_148_heading_222.7.jpg	0.0	0.0	42.2	38.8	0.0	19.0	0.0	0.0	0.0
building_149_heading_223.1.jpg	0.0	0.0	61.2	11.8	0.0	22.1	0.0	4.8	0.0
building_150_heading_223.1.jpg	0.0	28.4	46.4	11.1	0.0	14.2	0.0	0.0	0.0
building_151_heading_223.1.jpg	17.6	10.0	31.5	8.7	5.9	0.0	22.5	0.0	0.0
building_152_heading_223.1.jpg	0.0	4.8	14.2	15.6	3.8	25.6	13.8	22.1	0.0
building_153_heading_223.1.jpg	26.0	9.7	21.8	4.2	0.0	10.4	19.7	0.0	0.0
building_154_heading_223.1.jpg	14.5	12.8	33.2	11.1	8.0	7.3	13.1	0.0	0.0
building_155_heading_223.1.jpg	3.8	9.3	25.3	17.3	4.8	22.8	16.6	0.0	0.0
building_161_heading_223.1.jpg	2.8	0.0	73.0	4.2	0.0	20.1	0.0	0.0	0.0
building_168_heading_41.0.jpg	0.0	0.0	49.5	30.4	0.0	15.6	2.8	0.0	0.0
building_170_heading_41.0.jpg	0.0	0.0	22.5	18.0	0.0	30.4	11.4	0.0	0.0
building_171_heading_41.0.jpg	7.6	0.0	21.8	3.5	0.0	15.2	46.0	5.9	0.0
building_172_heading_41.0.jpg	23.2	0.0	23.5	9.7	0.0	22.8	20.8	0.0	0.0
building_173_heading_41.0.jpg	0.0	0.0	22.8	9.3	0.0	37.0	30.8	0.0	0.0
building_174_heading_41.0.jpg	0.0	0.0	22.8	9.3	0.0	37.0	30.8	0.0	0.0
building_219_heading_43.1.jpg	5.5	0.0	26.6	8.7	0.0	30.8	28.4	0.0	0.0
building_220_heading_43.1.jpg	7.6	0.0	9.0	28.4	0.0	31.1	23.9	0.0	0.0
building_221_heading_43.1.jpg	9.7	0.0	34.6	0.0	0.0	34.6	21.1	0.0	0.0
building_222_heading_43.1.jpg	9.3	0.0	35.6	0.0	0.0	19.4	35.6	0.0	0.0
building_223_heading_43.1.jpg	0.0	0.0	50.2	44.6	0.0	5.2	0.0	0.0	0.0
building_224_heading_43.1.jpg	8.7	0.0	23.2	68.2	0.0	0.0	0.0	0.0	0.0
building_225_heading_43.4.jpg	0.0	10.7	34.3	39.4	0.0	0.0	15.6	0.0	0.0



STF WORKSHOP
PROJECT

Project

Explorations + Q&A

Use this time to create a few images and ask any questions you may have. The images will be presented as final results of the STF workshops on Thursday 17th.

- Generate a few images using your models.
- Insert them into the designated slides within the presentation.

THANK YOU!

PRESENTERS |
IVANA PETRUSEVSKI - ESTHER RUBIO MADROÑAL - HESHAM SHAWQY - EVE TOBEY - CALLUM SYKES

GRIMSHAW
DESIGN
TECHNOLOGY